CHAPTER 12

WOOD-BASED PANELS: PARTICLEBOARD, FIBREBOARDS AND ORIENTED STRAND BOARD

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1. INTRODUCTION

Wood-based panels have become an important component in the wood processing chain. From a materials viewpoint wood-based panels provide a way of realising value from materials that are not suitable for other uses, or that are residue streams from other processes. However these panels are now an integral part of the wood products market, meeting requirements that were once met by solid wood. The ability to produce panels of larger sizes reduces the number of components needed while the uniformity of panel properties give a more consistent performance. The panels are used extensively in construction and in the furniture, cabinet and joinery industries, while the ability to 'engineer' these products to meet specific performance requirements has been a significant factor in their growth. The technology for producing these panels has developed significantly, in particular with the introduction of continuous pressing providing new levels of product uniformity.

2. OVERVIEW

Particleboard, (PB), medium density fibreboard (MDF) and oriented strand board (OSB) developed in the latter half of the 20th century from technologies first applied to plywood manufacture, but required new approaches to particle preparation and drying, faster setting adhesives and press design. Particles, being smaller than veneer sheets used to manufacture plywood, allow a much wider range of raw material to be used. In fact these panels use raw material that is either a waste stream from other processes or a wood resource that cannot be used for other purposes: increasingly wood from urban waste streams is being used, particularly for particleboard. Another precursor to these panels was the wet-formed fibreboard panel known as hardboard, where a fibre mat was pressed to a high density (950-1200 kg/m^3). This panel, also known as 'Masonite', was self-bonded: this being achieved by reaching a temperature in the hot press (205-215^{o}C) that caused thermal decomposition of the wood to produce an adhesive. Panel thickness was limited to about 6 mm by the necessity to achieve this temperature in the centre of the panel during pressing.

A wide range of adhesives are used. For the three panel types that are the focus of this discussion synthetic adhesives are used, although tannins extracted from bark have been used. However other binders such as gypsum and cement can be used with particles and with fibres. In this case the binder levels are much higher so that these products might be described as a wood-reinforced gypsum or cement matrix.

In general these products are pressed to consolidate the mat and to cure the thermoset adhesive. Whereas in plywood pressing the main object is to achieve good contact between the veneer layers with as little increase in density as possible, in the case of these products the average density is one of the key variables in determining panel properties and will normally be 30-100% greater than the bulk density of the wood from which the panels are made. A major advance in all these products is the ability to control the density profile through the thickness of the panel.

It is possible to characterise wood-based panels in terms of a plot of panel density against the major length dimension of the wood-elements used. In theory there is an infinite range of adhesive bonded products that can be defined within the range of particle size and density covered in Figure 12.1. In practice the range is limited by commercial requirements although the range indicated is by no means definitive, either for particle size or for density range.

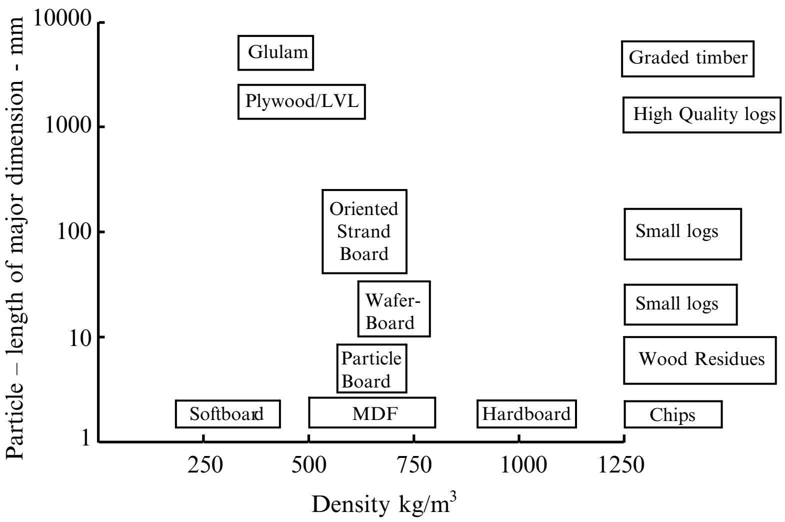

Figure 12.1. Classification of wood-based panels according to density of the panel and the size of the wood elements.

3. MARKET

Wood-based panels such as particleboard, MDF and OSB are a significant component in the wood industry. They satisfied performance requirements that are increasingly difficult to meet from solid wood sources and are able to meet these by using residues from other wood processing operations, or in the case of OSB, from small diameter logs from species that otherwise have limited potential, and at competitive prices. The technology to produce these panels has advanced to the point where these panels are made with a uniformity that allows processing on automated sawing, machining and finishing lines: increased automation of panel processing operations results in better quality components at a lower cost. Finishing systems range from paints and lacquers, grain printing, through to application of pre-printed or solid colour resin impregnated papers to provide wear resistance for working surfaces. To complete the cycle as it were, the composite panels can also be finished with a decorative veneer to provide a natural wood surface.

The market remains dynamic with newer panel types establishing markets by replacing other panel types. MDF grew initially in an area between particleboard and solid wood. The ability to create a machined or profiled edge enabled components to be manufactured from a single piece of MDF whereas particleboard required a wood clashing strip to provide such an edge. MDF also had much better ability to hold fastenings than did particleboard. This resulted in the introduction of new fixing systems for particleboard that overcame this limitation. Overall, particleboard has responded to the challenges posed by MDF by improving technology that raised its performance in a number of areas. Meanwhile the MDF sector has worked also to improve performance, not necessarily by increasing property levels, but rather by creating a more uniform product that performs more consistently. Lower panel densities (600 kg/m^3 as against 720-800 kg/m^3) have been used to increase plant capacity while maintaining performance and reducing costs.

OSB was developed explicitly as an alternative to plywood, at a time when that industry was faced with falling availability of modestly priced, large diameter logs for traditional construction plywood. OSB capacity has increased rapidly and now provides a competitive alternative to plywood sheathing over the timber frame of a North American house. While most OSB production is in North America, capacity is increasing elsewhere, particularly in Europe but also in South America. While initial growth was as a replacement for plywood, it has evolved to fill new market niches.

Panel volume data is available from several sources. While the FAO yearbooks provide some data, industry publications offer a more up to date perspective. The data are not always consistent or comprehensive. FAO data include OSB within the particleboard classification. Further FAO data cover world wood-panel production volumes whereas industry surveys (Wadsworth, 2005) are based on plant capacity of all operational units in each region. The difference between these two numbers reflects capacity utilization which varies as demand rises and falls. The volumes in this analysis are used to indicate trends rather than absolute levels (Table 12.1).

Table 12.1. World production of particleboard, OSB and MDF for 2004.

	million m^3	%	Average growth, 1995-2005 million m^3/yr
Particleboard	81.5	54.8	2.4
OSB	26.5	17.8	2.1
MDF	40.7	27.4	3.5
Total	148.6		

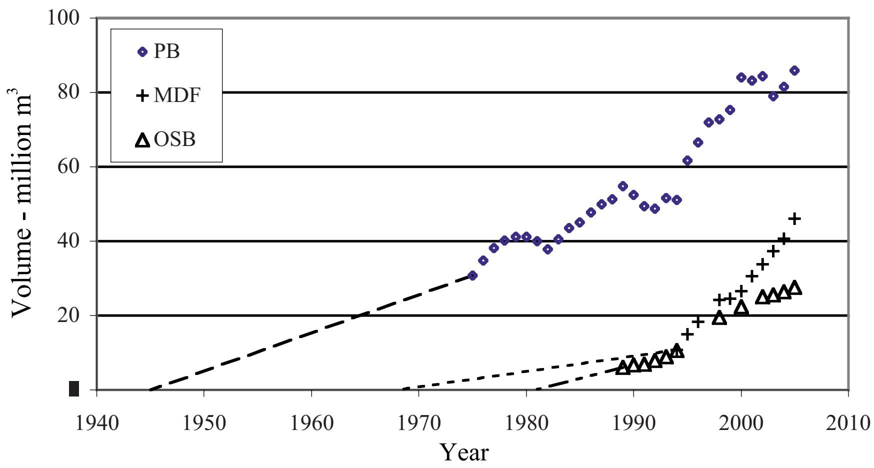

Figure 12.2. Annual world volume capacity for particleboard, MDF and OSB. The data are extrapolated back to the year when the products were first produced commercially.

These are comparatively new products with these volumes having been achieved in a relatively short time. The first particleboard was produced in 1941, with commercial production starting in the late 1940s. MDF was produced in 1965, with the first plants in USA in the early 1970s. Both particleboard and MDF moved quickly beyond the country of origin as the benefits of the new panels were realised. Commercial OSB production began in North America in 1981. World capacity is shown in Figure 12.2.

Product replacement is significant where two types of products compete within the same market area:

- Oriented strand board has grown as an alternative to plywood. For North America in 1987 OSB production was 29% of the plywood volume; by 1994 this had increased to 55%.
- Similarly in 1995 world MDF capacity was 24% of particleboard volume; by 2005 MDF had grown faster to reach 54% of particleboard volume.
- A greater proportion of MDF (43%) was exported from the country of manufacture over the period 1998-2002 than for particleboard (27%) although this varies significantly between different countries. Prices for particleboard are lower than for MDF by some 15-20% making this product more sensitive to freight costs, and therefore less likely to be sold in export markets at an acceptable price.

Growth has been sustained: annual increments over recent selected periods are shown in Table 12.1. The growth rate of MDF includes a major increment in China that may include some capacity from plants installed earlier.

A 1998 survey of manufacturing costs in the Canadian industry provides comparative figures for these panels. Table 12.2 shows the price advantage that has driven the use of OSB as a replacement for plywood, once satisfactory performance had been demonstrated. Waferboard was developed in 1962, but OSB had to await the optimization of flake geometry and the development of methods allowing the flakes to be oriented. Only then, in 1981, was OSB able to demonstrate a level of performance that allowed its use as an alternative to plywood.

Table 12.2. Canadian panel manufacturing costs (Poliquin, 1998). Note the relatively high costs for wood and labour in a plywood plant.

Cost $C/m^3	Softwood plywood (BC)	OSB (Quebec and Ontario)	Particleboard	MDF
Wood, net cost	151	58	28	40
Adhesive	17	23	35	36
Wax	0	4	3	4
Labour	103	29	22	20
Electricity	11	8	10	14
Misc.	22	18	17	22
Total Variable	304	140	115	136

The survey reinforces the benefits of large capacity automated plants producing OSB, particleboard and MDF have over smaller more labour intensive plywood plants. There are significant benefits of scale. These are achieved firstly by maximizing capacity of an individual press line, secondly by adding production lines in existing facilities. However increasing line capacity gives the greatest benefit, at least for new plants. Particleboard line capacities range up to 1800 m^3/day, while MDF press lines range up to 1500 m^3/day. The average line capacity for an OSB plant in North America is 1100 m^3/day, while new OSB lines proposed for 2005-6 have an average capacity of 1600 m^3/day.

4. CHARACTERISING WOOD-BASED PANELS

Wood-based panels such as particleboard, oriented strand board and medium density fibreboard consist of particles of widely varying shape and size bonded together with an adhesive system.

4.1. Wood particles

The variable shapes and sizes of the wood elements contribute significantly to the particular properties in any panel and in the way it can be used. These range from veneer sheets for plywood and LVL (Chapter 11), to individual fibres and fibre fragments used in fibreboards. The shape of the particles has become increasingly important as producers seek to maximize product properties while minimizing cost – for good structural properties long, thin particles and fibres are desired. The preparation of particles of the desired geometry and the separation of these from particles that do not conform is of increasing significance in quality control. Here, for convenience the term particle may refer to particles (chips, flakes and wafers) as well as fibre (fibres and fibre bundles).

4.2. Adhesive

The adhesive is selected to meet the specific strength and durability performance requirements of the panel, but also it has to meet other requirements. Generally the adhesives are thermosetting, in that they undergo permanent change with the application of heat. Cost is a major consideration, with the cost in the panel being determined by both the purchase price of the resin and its usage rate. An adhesive that requires higher temperatures to cure will also increase the cost of the panel by requiring longer times in the hot press.

Over the last 20 years, environmental concerns have become an important consideration in adhesive formulation and use. Firstly, in the plant the adhesive may require particular handling or the use of protective equipment. Secondly, volatile emissions arising from adhesive reactions both in the hot press and subsequently when the panels are in service are subject to tight regulatory control. In particular the reduction formaldehyde emissions from wood-based panels has been a major objective in adhesive development over the last few years, both because formaldehyde-based adhesives are the major type used and because these have been implicated in environmental and health concerns.

4.3. Pressing

The pressing of the particles – after being mixed with the adhesive – into a panel is the point where the composite panel is formed. Conceptually the press provides the desired clamping force that is used to increase the contact areas between the individual surfaces being glued, while heating the material in the press reduces the time necessary for the adhesive to develop sufficient strength. The development of panel presses has allowed the overall density of the composite panel to be increased, often significantly above the density of the wood from which the particles are prepared. This ability to change the density of the product has been further refined to the point where the density distribution through the thickness of the panel can be controlled.

4.4. Density

For plywood and LVL the objective in pressing the panel is to create only enough pressure necessary to form good bonds. However, the average density of the panel in the cases of OSB, particleboard and MDF is well above that of the wood from which the particles are derived. In the case of these panels, strength properties are as much a function of the resin system and the manufacturing processes as they are of the wood being incorporated into the panel.

However to generate good bonding through greater interfacial contact between particles and fibres it is necessary to densify the mat in the press. High density woods require greater force to form bonds of adequate strength and in turn produce undesirably heavy wood panels, a feature that can be countered by adding more adhesive and reducing the compression on the mat. The market prefers medium density woods (400-550 kg/m^3). It follows from beam theory (Chapter 10) that the reduction in thickness that results from increasing the density of the material in the hot press has a greater effect on the bending strength of the composite than does any improvement in bond strength. For OSB made from aspen/poplar mix the panel density ranges from 576-640 kg/m^3. When made from denser southern yellow pine, the common wood for U.S. plants, the panel density range is 608-672 kg/m^3.

The density of the panel is important in determining its properties, as is the ability to control the density distribution within the thickness of the panel. The bending strength of MDF depends on the tensile strength of the outer layers. Therefore increasing the density in the faces of the panel allows the bending strength to be improved. Concurrently a lower density core cuts fibre consumption compared to that that would be required if the core density were close to that of the face of the panel. Controlling the distribution of density through the thickness of the panel allows strength properties to be maintained at a lower overall panel weight.

4.5. Mat structure and lay-up

The method of assembly of the particles ahead of the press can also be used to manipulate the properties of the composite. For example, it is possible to use different particles (or adhesive systems) in different layers of the panel. Indeed, this approach allows composites with totally different structures in different layers of the panel to be produced. Such a product is 'Triboard' produced by Juken Nissho in New Zealand, where a product with MDF faces and a strand core is produced in a single pressing operation.

In a less dramatic form, separation of the fine and coarse particles in the mix prepared for particleboard during the mat forming process can produce panels with very different performance objectives. If the larger particles are concentrated in the outer surfaces of the panel, the bending strength of the panel will be improved (for flooring and structural uses). If the mat forming is configured to concentrate the fine particles in the outer surface of the panel, the surface will be much smoother, and respond better to paint and lacquer finishing systems, but the bending strength will be reduced. This approach is used where the panels are to be used for furniture or decorative face panels in joinery.

Different adhesives can be incorporated in different layers of the panel. The use of a faster setting resin in the centre of the mat will reduce the time needed to complete the resin cure in the hot press. The faster setting resin may be more expensive, or have other disadvantages, but these may be offset by using a slower curing, less expensive adhesive in the outer layers of the mat. This increases the complexity of the plant with separate lines needed to prepare material with different resin types or characteristics. This expense must be justified in terms of the increased production rates achieved in the press, in resin cost savings, or in improved product properties.

The lay-up or mat forming stage offers another opportunity to influence the directional properties of the panel. It is possible to orient the particles in one direction of the panel. A parallel alignment of members is inherent in glue laminated beams, while alternating the grain direction of veneers as they are laid up to prepare a plywood sheet for pressing is a way of achieving more uniform properties in the plane of the panel. While orientation is relatively straightforward for beams and veneer sheets, the technique has been applied also to smaller particles. As suggested by its name, orientation of the particles is important in manufacture of oriented strand board (OSB).

Orientation becomes more difficult as particles become smaller and consequently increase greatly in number. Some orientation does occur, even with the fibre used for MDF. In forming the mat the fibres are deposited on a moving belt that results in a preferential degree of orientation in the direction that the belt is moving. This effect shows in the bending strength of MDF which is greater in the direction of the mat travel than perpendicular to this direction. The difference depends of the speed of the forming belt and can be 20-25% higher in the belt direction than at right angles to this.

The shape of the space within the press influences the final form of the panel. For most applications a flat panel is preferred and the formed mat is pressed between flat platens. However there are plants using platen pairs formed as a three-dimensional die to produce a composite structure shaped for a particular application. The most successful of these 3-D shapes is the door skin, where the textured die is configured to produce the face of a wood-panelled door. The door is assembled by gluing the two faces to stiles and rails and trimmed to produce a finished product at a cost that has allowed this product to dominate the market for internal doors in buildings. Other successful three-dimensional wood-based composite products are shipping pallets, and circular table tops. The requirement is for a large volume. A door skin press may have 20 die sets, each able to produce the two matching faces of the door. If the total pressing and unload/loading time is 60 seconds, the press produces panels for 20 doors per minute, or in excess of 25 000 per day assuming the press line runs for 90% of a 24 hour day.

Early in the history of particleboard manufacture, extrusion was used to make profiled shaped material, but this process never became significant even where such a process might be expected to have advantages, e.g. for a product with an irregular cross-section and required in long lengths such as interior finishing mouldings. Instead, these are produced in large quantities, from MDF, not by an extrusion process, but are machined from strips sawn from flat panels. Extrusion has come back into fashion in the production of wood-plastic composites where the ratio of wood to polymer is close to 50:50 (Wolcott, 1996).

5. HISTORY OF WOOD-BASED COMPOSITES.

The range of wood-based composite panels available in the market is a result of the continuing development of processing techniques and adhesive technologies (Table 12.3). Wood glues have been used for centuries, but availability in quantity with consistency of performance was achieved only when synthetic formaldehyde-based adhesives became available.

Table 12.3. Evolutionary sequence for wood-based panels (Clark, 1991).

Year	Product	Country
1830	Mechanically sliced veneer	France
1896	Rotary peeled veneer	Estonia
1898	Fibreboard (wet formed)	UK
1906	Plywood	USA
1914	Insulating board	Germany
1925	Hardboard	USA
1941	Particleboard	Germany
1945	Dry process fibreboard	USA
1966	MDF	USA
1969	OSB	Germany

Plywood was originally produced using natural adhesives and later with phenol formaldehyde (PF), discovered by Dr Leo Baekeland in 1906. The first fibreboard panels were produced using papermaking technology. In this case the wet formed mat was made using a relatively crude fibre, and dried to form a low density panel called Softboard with a relatively low strength that limited its application. Subsequently this low density panel was pressed to a much higher density at platen temperatures that caused thermal decomposition of the wood to commence. This resulted in the panel known as hardboard. Hardboard is significant in that it was the first panel product made from fibres and fibre bundles to demonstrate property levels approaching those of plywood.

It was not until after World War II that particleboard appeared as the first product to use urea formaldehyde (UF) as an adhesive. Particleboard provided a low cost way of converting low grade raw materials into a useful product. Capacity increased rapidly after the World War II to meet the huge demand for building materials, especially in Europe and the technology quickly spread to other countries Experience with this product showed the benefits of controlling the particle size and geometry and lead to improved performance.

Early fibreboards were developed by taking fibre from a wet-process fibre line, drying this, mixing it with resin, and pressing it as had been done for particleboard. However, in the last three decades medium density fibreboard (MDF), a UF bonded and dry formed panel, has largely replaced the early fibreboard products.

Oriented strand board (OSB) was developed as an alternative to plywood and is now used in large volumes in construction, particularly in North America. It uses small thin flakes or strands which are oriented to provide directional properties.

These are prepared with specialist flaking equipment that can use small diameter logs. Initially this was from species having no other use, but as capacity has grown alternative wood sources have been used. The process evolved from the waferboard manufacture that used thin flakes or wafers to form a panel where the wafers performed as small veneers. The increased slenderness ratios of strands resulted in improvements in both strength and directional properties so that OSB became an alternative to plywood. The reduced width of the strands compared with wafers allowed them to be produced from smaller diameter wood.

The development of the wood-panel industry from 1966 has been chronicled in the proceedings of the annual Washington State University Symposia on Particleboard and Composites.

6. RAW MATERIALS

Wood-based composites require two major raw materials. These are:

- Particles, which are generally produced from wood although other cellulosic materials have been used, with straw being used in Europe early in the development of composites.
- An adhesive system that is compatible with the material to be bonded.

Other minor components may be included to impart specific properties, for example the addition of wax to improve short term response to wetting is almost universal. Insecticide and fungicides to impart resistance to fungal or insect attack may be added, while fire retardant materials can be incorporated to give panels with a specified fire performance.

6.1. Chips, fibres and sawdust

Each panel type requires particles of a particular quality and configuration. Veneer-based products such as plywood and LVL, and components for glue-lamination require the use of high quality logs or timber. The strands for OSB are engineered flakes that are prepared from logs. The flaking process is able to use logs that are too small for sawmilling, increasing the proportion of the tree that can be used, or alternatively using species such as alder and poplar that are unsuited to other uses.

Particleboard utilizes residue materials from other wood processing operations. Sawdust and shavings predominate, although most plants are able to use chips or flakes made from roundwood to meet specific strength requirements in the product. Cost is the major driver, with the proportion of high cost material such as chips or roundwood limited to that necessary to meet product requirements. Particle size distribution and shape have become more important as the effect of these variables on product quality has been recognised, leading to reductions in cost.

MDF is also produced from chips, but increasingly sawdust and shavings are used as fibre sources. As the name suggests the particles in this case are fibres or fibre bundles characterised by a high slenderness ratio, typically 80-100:1 for whole individual fibres. Most of the fibres in sawdust and shavings are damaged in the processes that lead to these residue materials, so that their average fibre length is reduced to the point where it is more difficult to meet strength requirements where these feed materials are used. Agricultural residues such as rice or wheat straw can also be used for both particleboard and MDF. However they often contain components such as waxes that require particular treatment, or the use of specific adhesives to achieve satisfactory bonding. Recently, reclaimed wood from urban waste streams has been included in the furnish for both particleboard and MDF.

Particleboard and MDF plants are thus tied closely to residue streams from other wood processing operations. The logs used for OSB may come from a specific small diameter resource or from top logs arising from harvesting of other forest resources. The processing of these residue materials into particles of the required shape and thickness is specific to each raw material.

6.2. Adhesive systems

A wide range of adhesive and other binder systems is used in the manufacture of composite panels, dominated by formaldehyde-based systems. Growth in the composite panels industry has depended on adhesives with the required properties being available in sufficient quantities at a suitable cost. The industry requires considerable economies of scale for synthetic adhesive production. The materials are derived from oil (phenol) or natural gas (urea and formaldehyde). Recently in the face of environmental constraints and the increasing cost of oil and natural gas there is renewed interest in adhesive systems derived from renewable resources.

The earliest wood adhesive for mass production of panels was phenol formaldehyde (PF). This is widely used for products designed to perform in severe weathering conditions and dominates the exterior plywood market. Resorcinol formaldehyde is a somewhat similar resin system that cures at ambient temperatures, but the cost is much higher, limiting its use in high-volume applications. These systems, although formaldehyde-based, do not have the continuing formaldehyde emission problem that is associated with other formaldehyde-based systems.

This other class of formaldehyde-based adhesives is the amine-based systems where the formaldehyde is reacted with urea (U) or with melamine (M).

The urea-based systems (UF) are the most commonly used and cure at relatively low temperatures (*c.*105°C), reducing time in the press. The glue bonds are less durable so these adhesives are used in products designed for applications where extreme weathering is not a consideration. The one problem with UF adhesives is that there is continuing emission of formaldehyde from the composite over the first year or so of its life. The UF polymer formed in the press can break down to formaldehyde and urea, at a rate that increases with temperature and moisture content. The formaldehyde lost during the pressing of UF bonded panel arises from this breakdown of the UF in the outer layers of the panel that reach temperatures close to that of the hot platen. Continuing formaldehyde emissions from the panel in service can affect the health of people working in built environments – causing various allergic reactions. A large reduction of formaldehyde emissions has been and continues to be a major development objective for UF resin manufacturers, with significant progress having been made. Formaldehyde emission classifications are specified in standards and are very often regulated by specifying the required emission classification. Roffael (1993) provides a useful review of UF resins and the factors influencing formaldehyde emissions, while Young (2005) provides a recent review of the tests used to measure formaldehyde emission levels and correlations between these.

Melamine also reacts with formaldehyde to form an adhesive system (MF). While the UF polymer is largely linear, the cyclic nature of the melamine molecule with three amine groups allows a cross-linked structure to develop. Melamine is 4-5 times as expensive as urea, and pure MF is not commonly used in composite panels. An attractive feature of MF resin is its miscibility with urea formaldehyde, allowing resins containing both melamine and urea formaldehyde components. In replacing a portion of the urea, one can achieve the level of cross-linking appropriate to the end use requirement. Low levels of melamine (2-3%) are incorporated in UF resins to improve strength properties for panels using UF resins with low formaldehyde emission characteristics. Higher levels (10-15% melamine) are used to improve the moisture resistance of the panels. The general term MUF is used for both these systems, but an indication of the melamine level is necessary to describe the performance attributes.

Three component formaldehyde systems have also been developed. In addition to melamine and urea these also incorporate phenol and are known as PMUF. The levels of each component can be varied to meet the specific resin performance required at minimum cost.

Tannins derived from tree bark also react with formaldehyde to form durable adhesive bonds. Tannins have been widely investigated, and the subject is well covered in the literature (Pizzi, 1989). The dark colour of the resin limits applications to those where a light colour is not important: indeed substrate colour can show through some overlay materials. Tannins are used commercially in a number of particleboard products notably a structural flooring particleboard panel used in Australia. Compared with the synthetic adhesives, the tannins, together with other adhesive systems derived from natural materials, are much more variable. This has limited their use, with the increased processing necessary to achieve uniformity increasing the cost to the point where there is no advantage. Other adhesive systems based on chemically modified plant derived raw materials such as oils have been proposed.

Isocyanate systems based on polymethylenediisocyanate (PMDI) are being used increasingly. This adhesive has a completely different chemistry to formaldehyde-based systems. While expensive, the absence of formaldehyde emissions is a determining factor in some applications. An attractive feature of isocyanates is the excellent wetting and coverage of the wood surface so requiring lower usage rates. Early on one problem with isocyanates was their ability to stick to the hot platens – a problem solved by use of release agents. Initially these were sprayed on the platens to prevent adhesion, but are now incorporated in the resin system.

PF and PMDI are classified as suitable for exterior rated panels while MUF (at the higher melamine level) is regarded as suitable for service in interior situations where some level of humidity or wetting may occur (bathrooms, kitchens, roof linings). UF resins are generally regarded as suitable for interior applications where contact with water would be unlikely. In addition to the resin system, many other factors such as particle size are significant in panel performance. Durability is discussed in more detail in a subsequent section.

Inorganic binder systems are also used. Cement bonded products both as particleboard and as fibreboard are used in exterior situations with the added advantage that these products resist fire. Generally binder levels in this case are quite high (*c.* 35% wt/wt basis), and the product can be regarded as a particle or fibre reinforced cement board. Gypsum is also used – both particleboard and fibreboard being manufactured with this binder. The inorganic nature of these binders gives panels with improved resistance to fire. Gypsum has the advantage of being a low cost residual byproduct created in the cleaning up of flue gases in coal-fired power plants. Indeed unreinforced gypsum board with paper overlays is the major contender with wood-panels for non-structural wall linings.

Chemical compatibility between the resin and the wood can also be an issue. UF and MUF systems require an acid pH environment (generally <5) to cure the resin at a reasonable rate. The acidity of wood species varies over a wide range and can affect resin cure speed. Acid generating catalysts are generally used in particleboard, these are less effective in MDF where the preheating of the wood prior to separating the fibres generates further acid groups on the fibres (Murton, 1998), and where the much greater surface area of the fibre allows the buffering capacity of the wood to determine the resin cure rate. This aspect is also an issue in cement bonded products where an alkaline environment is necessary to allow the cement to bond with the wood surface.

Within this wide range there are many possibilities. The adhesive purchase price and the required addition rate are major factors in determining the overall adhesive cost for a panel (Table 12.3). Performance requirements, particularly durability may limit the available choice. Overall UF systems provide the lowest cost solution for MDF and particleboard and are used in the greatest proportion of these panels, in excess of 95% for MDF.

7. GENERALISED PANEL PRODUCTION LINE

The production of composite panels (Figures 12.3 and 12.4) is based on the following stages that generally, but not exclusively, proceed in the following sequence:

- Preparation of the particles.
- Drying to the required moisture content.
- Spreading the adhesive.
- Forming the mat for pressing.
- Pressing the panel.
- Finishing the panel: sanding, trimming *etc*.

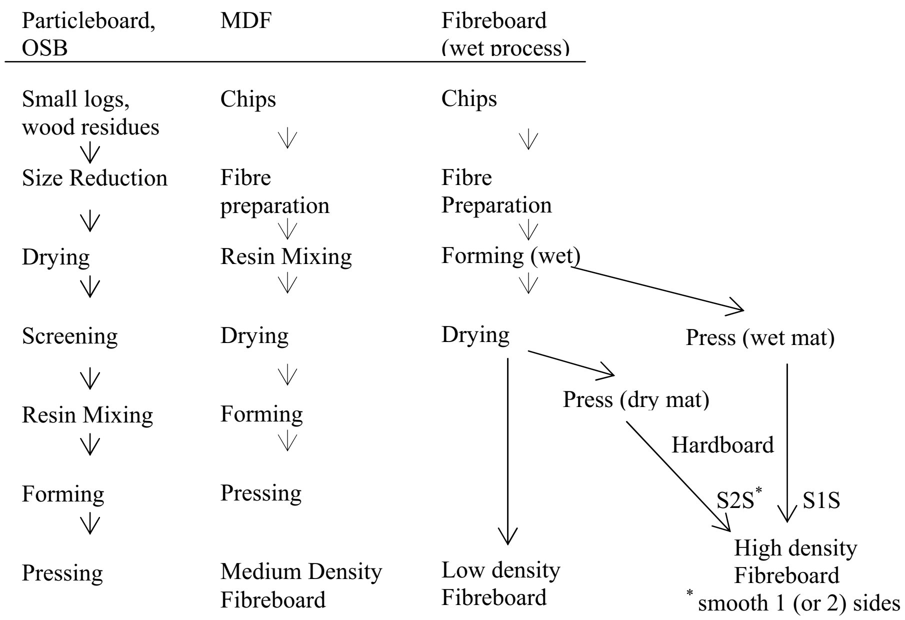

Figure 12.3. Production steps for particleboard/OSB, MDF and fibreboard (wet process).

7.1. Particle preparation

At first, particleboard was made with minimal preparation of the material, often using planer shavings and sawdust directly. As experience was gained, recognition of the importance of particle geometry and size in determining particular performance properties lead to much more attention to these aspects of wood preparation. Today, as particleboard producers have sought to maintain their market against competitors producing the same product and newer panel types, particularly MDF the size distribution and shape of the particles is closely controlled by a range of size reduction processes and size-based separation techniques. The process requirements are determined by the starting material and the particular product requirements. Broadly the size reduction process can be divided into two categories.

7.1.1. Impact mills

Impact mills fracture the material by striking it with a hammer, which is usually rotating at high speed. This approach induces a random failure in the material that probably starts near the impact point and continues either in the general direction of the stress induced by the impact or along the line of a weakness in the material effectively shearing parallel to the grain. The randomness of the process means that neither the alignment of the particle nor its position relative to the impact zone is controlled. The result of an impact operation is therefore an overall reduction in size

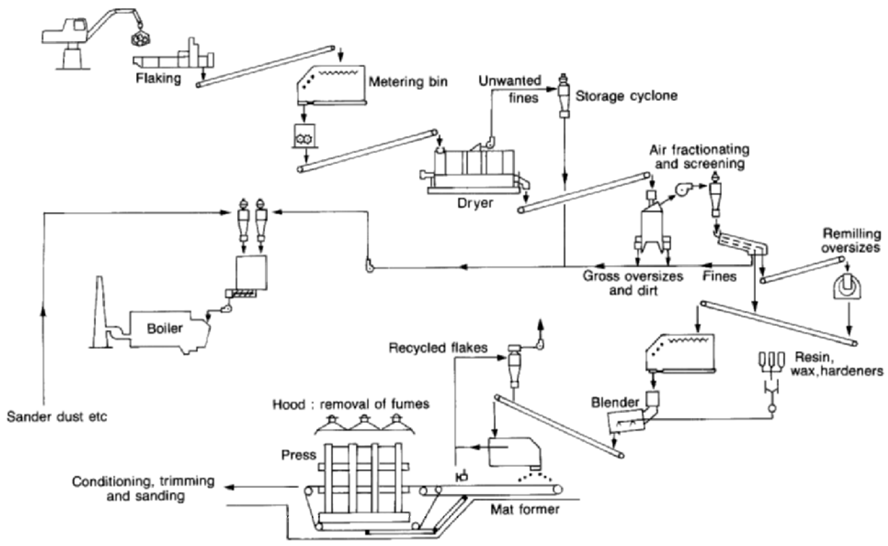

Figure 12.4. Material flows for the manufacture of OSB (Courtesy Siemplekamp, Germany).

of the incoming particles into a number of smaller particles with broad distribution of particle sizes that is determined by the impact energy and the original geometry and fracture characteristics of the material.

Impact devices (Figure 12.5) thus represent a relatively crude approach to size reduction. They are acceptable for processes where a range of particle sizes is required, perhaps to give a smooth surface as well as a coarser core layer. The main advantage is that they require little maintenance and are less susceptible to damage by foreign inclusions (metal, stones). However the random nature of the impact results in a wide range of particle sizes in the discharge some of which may be too small for the product. These machines are usually followed by a screening operation to separate oversize material (that may be processed a second time) and fines from the acceptable particle size range.

7.1.2. Engineered particles and flakes

The other approach to size reduction has been to use a knife to cut the material in a situation where the material is presented to the knife in a controlled manner so that the orientation of the cut and the distance between the cuts is controlled (Figure 12.6). Such devices include veneer slicers (which are not used in the preparation of particleboard furnish), chippers and flakers. These machines rely on close control of the relationship between the cutting knife, the backing plate that holds the wood in relation to the knife and the angle of presentation of the wood.

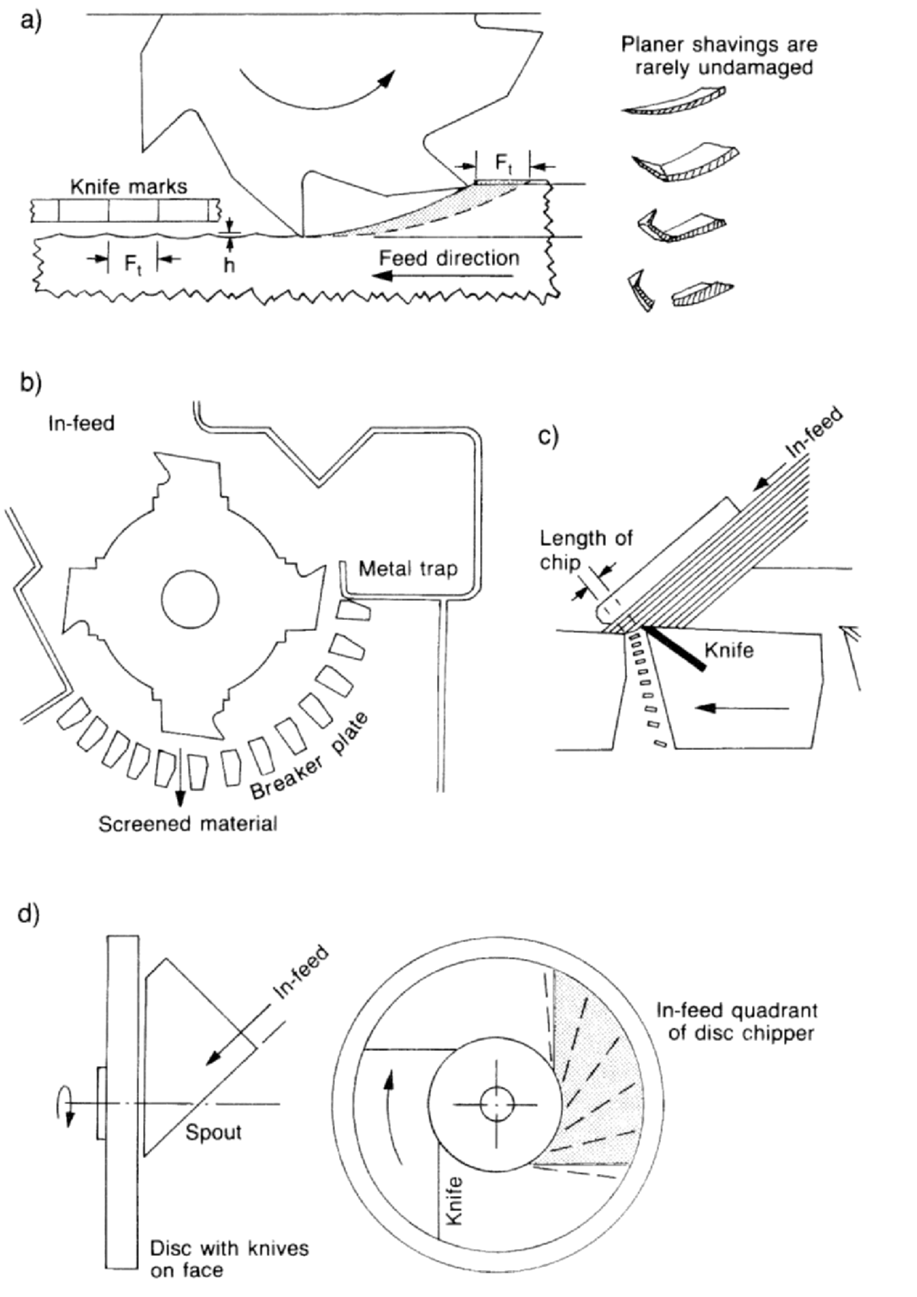

Figure 12.5. The generation of wood chips (Koch, 1964). (a) Planermills produce shavings of variable geometry. (b) A hammer hog reduces coarse material. (c) The cutting action of chipping knives. (d). A disc chipper.

The ring flaker is an example of this type of machine. It is designed to produce flakes for particleboard from chips. The knives are placed in an outer ring facing inwards. The projection of the knives beyond the backing plate of the knife ring determines the thickness of the flakes. A counter-rotating inner rotor with bars controls the presentation of the chips to the knives such that it both aligns the chip flat against the backing plate on the knife ring as well as providing the force necessary to move the chip past the knife. The distance between inner bars on the rotor and the knives must be less than the design flake thickness to maintain this for the last flake to be taken from the chip.

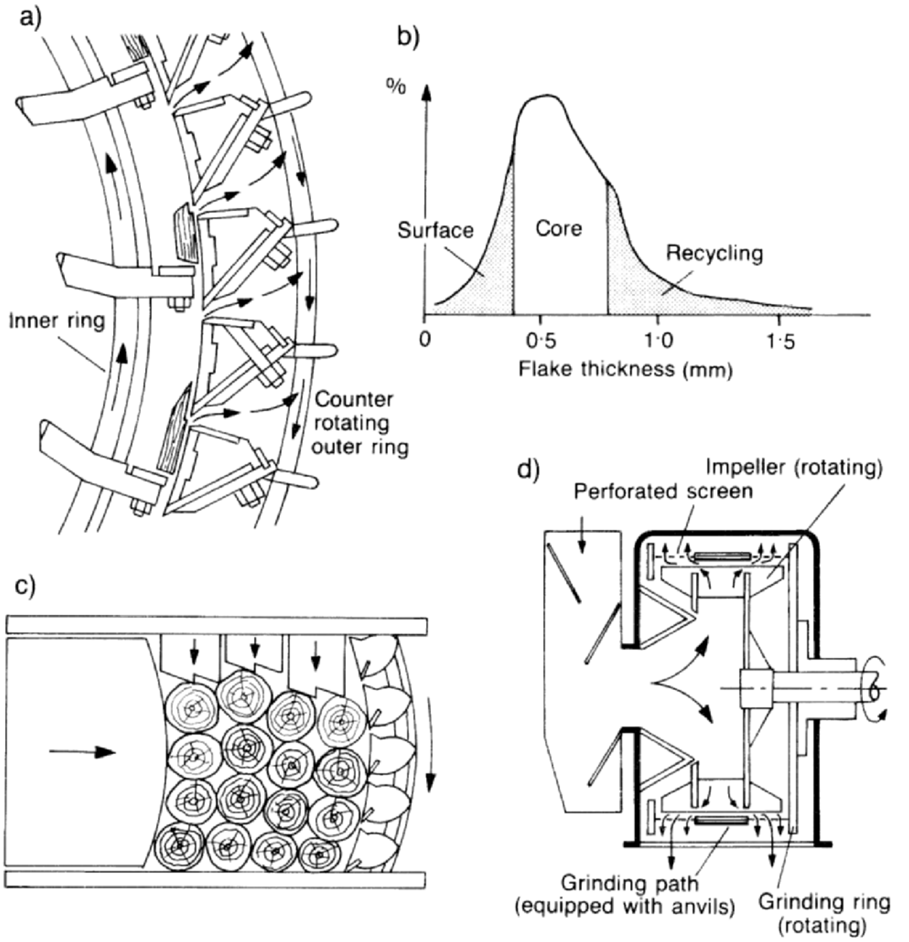

Figure 12.6. The generation of flakes. (a) Cutting action of a knife-ring flaker (Fisher, 1974). (b) Typical distribution of flake thickness (Jager, 1975) (c) Cutting action of a large drum flaker (courtesy Pallmann, Zweibrucken, Germany). (d) Ring refiner (Fisher, 1972).

Particleboard is unique among wood-based panels in that a range of particle sizes is desired. The requirement for particleboard varies, from applications where surface smoothness is the primary requirement to panels where strength requirements are more important. In the first case a range of particle sizes are needed to fill the gaps between the larger particles, to create a smooth surface. Where strength is the primary requirement then the material will be selected by rejecting the majority of the smaller particles at least from the surface of the panel that contributes most to the bending strength. The variable chip quality, arising from utilizing low grade wood resources in conjunction with a hammer mill, can be addressed by screening operations that separate oversize material for reprocessing and undersize particles for use in another part of the panel or as fuel. Screening leads to improved property values while at the same time reducing the amount of resin required. Engineered flakes prepared from chips are often used where improved strength is required, despite the much greater complexity and maintenance requirement of the ring flaker.

The preparation of fibre for MDF uses an impact device (the refiner) to separate the fibres but this follows a thermal pretreatment that softens the lignin. Consequently separation occurs in the lignin-rich middle lamella between the fibres so preserving as much of the cellulosic fibre structure as possible. The preparation of MDF fibre is considered later.

Stranding machines for OSB strand preparation use a similar geometry either in the form of knives placed radially on a disk, or in a cylindrical configuration similar to that ring flaker discussed above. With the input material in log form the force necessary to maintain the relationship between log and knife can be generated by mechanical clamps and pushers. This configuration will result in a small sheet of veneer with a length equal to that of the knife, or otherwise determined by circumferential scoring blades that cut across the grain ahead of the each knife. The path of the veneer sheet as it leaves the knife determines the width of the strand. A sharp change in direction a fixed distance after the knife will cause a fracture along the grain direction, determining the strand width. The stranding machine produces a flake or a strand with all three dimensions determined by the relationship between the knife, the support plate and the path available to the flake as it leaves the knife. Strands produced at the start and end of each log, or at the start or end of each cut may not reach the required quality level. These are separated by screening to be used in the core of the OSB panel, or used as fuel.

7.2. Drying

The approach to drying is constrained by particle size: the larger the particle the longer it takes to dry. Particles, flakes and strands typically pass through a rotating drum dryer, where the particles are lifted by projections (flights) on the drum surface to fall and tumble repeatedly through a hot gas stream. The drum may be arranged so that the particles have several passes, back and forth along the drum (Figure 12.7). The hot gas is usually obtained from a direct combustion source, and flows co-currently with the particles. Inlet gas temperatures can be quite high (up to 850°C) with the burner close to the dryer. The lightest particles are carried rapidly through the dryer in the combustion gas stream before being separated from this in cyclones. Larger particles such as the strands for OSB are too large and heavy to be pneumatically carried and tumble through the dryer to be discharged onto a conveyor: these particles remain in the dryer for up to 5 minutes. The exit temperature of the dryer is typically 105–120°C.

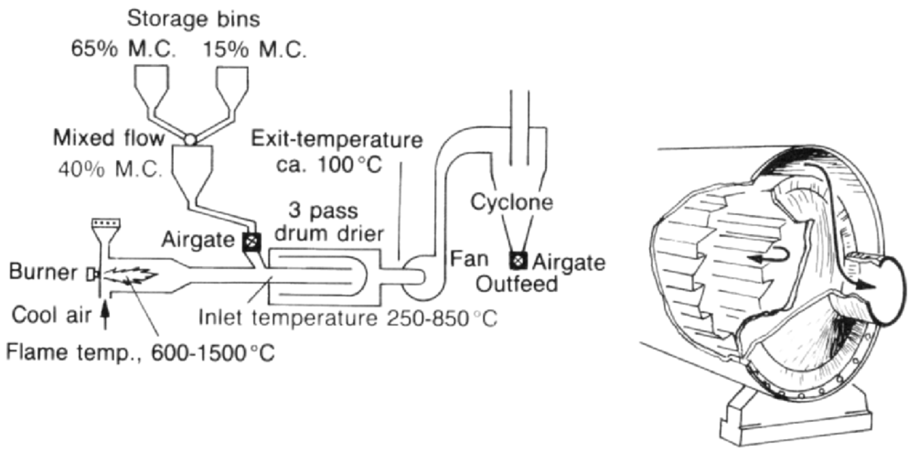

Figure 12.7. A three-pass rotary dryer that is often used in particleboard mills.

The high temperature used in particle dryers causes some breakdown of the wood. Although the evaporating moisture on the particle surface means the wood does not experience the high initial gas temperatures, this increases as drying proceeds. The exit temperature of the dryer is typically 105-120°C. The dryer gas flow is usually treated to remove particulate material as well as volatiles from the wood before discharge to the atmosphere. A range of treatment options are available with a wet electrostatic precipitator being commonly used.

The final moisture content of the particles leaving the dryer is in the range 2-5%. This is determined by the moisture content required in the press which is made up of that in the particles after drying and the moisture added with the resin.

Drying fibre for MDF takes a very different approach. The drying time for individual fibres is much shorter (<<5 seconds), allowing a flash tube dryer to be used. The fibre enters the stream of hot air that carries it to a cyclone that separates the dry fibre from the air. This short drying time has allowed the resin to be added before the dryer: the so-called blowline blending process overcomes the difficulties of mixing fibre and resin in a mechanical blender. The short drying time also changes the dynamics of the dryer, with the fibre being in equilibrium with the humid air stream that leaves the dryer. With the resin also travelling through the dryer, the moisture content of the material leaving the dryer is that required for the mat entering the press, typically 9-12%.

7.3. Resin blending: particleboard and OSB

Mixing the adhesive with the particles is critical in maximizing the resin performance and the approach again depends on the particle size. The aim is to have every particle contribute to the strength of the composite so that the resin particles need to be as small as possible and distributed uniformly over the surface. On contacting the dry particles moisture passes from the resin to the particles so the viscosity of the attached resin droplet increases.

The strands used for OSB are large enough to collect the resin particles – both solid and liquid resins are used for OSB – as they fall through a cloud of resin particles. The resin blenders used in OSB are large drums, typically 3 m or so in diameter rotating about a horizontal axis (Figure 12.8). Lifting bars carry the particles up to the point where they fall through the resin particles produced by the

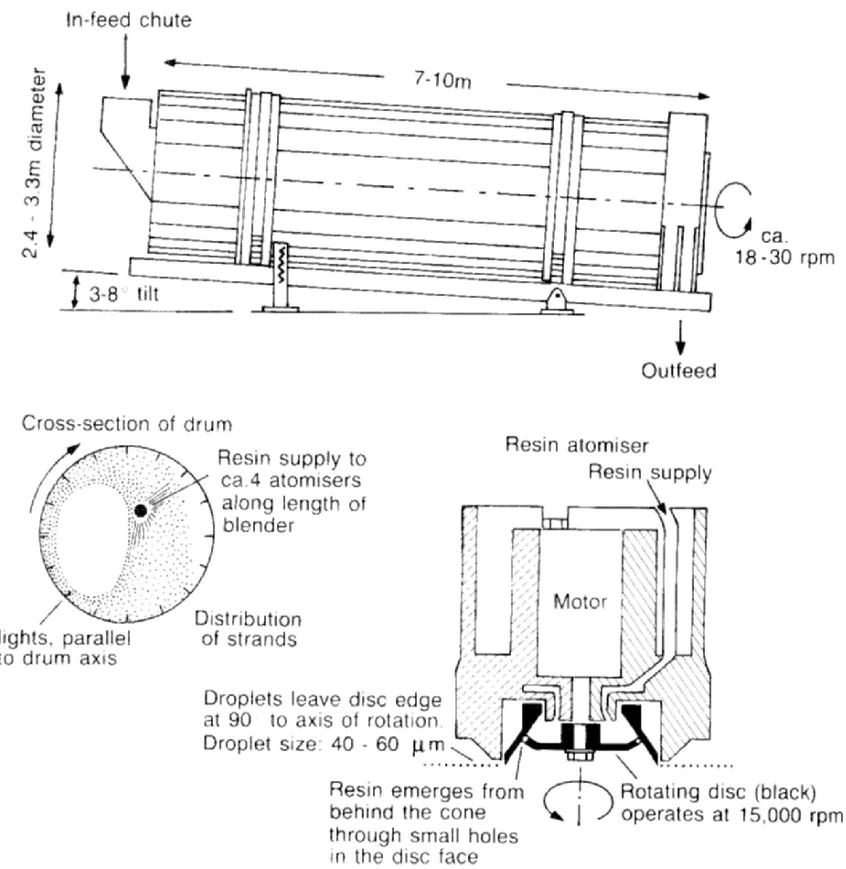

Figure 12.8. A long residence time blender (courtesy Coil Ind, Vancouver, BC).

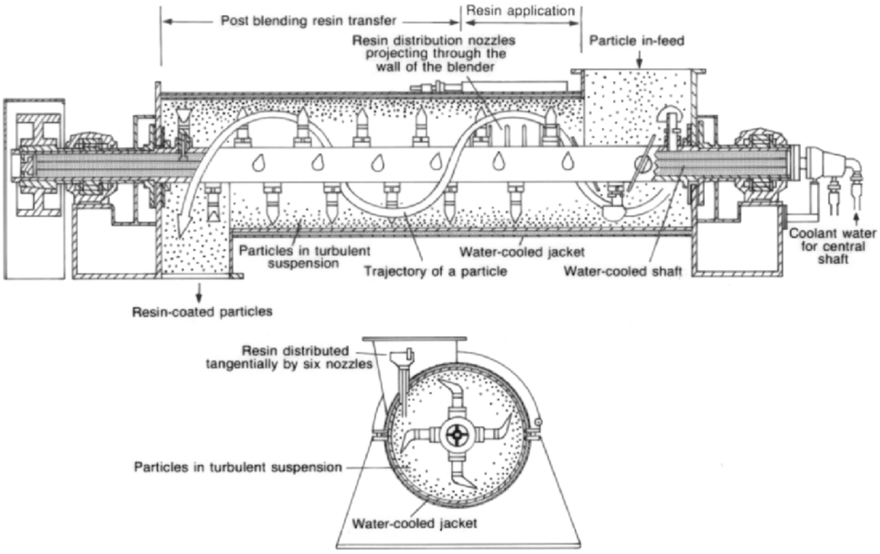

Figure 12.9. A short retention time blender (courtesy Draiswerke, Allendale, NJ).

nozzles. Solid resin is blown into the drum, while liquid resin is atomized in spinning disk atomizers installed along the length of the drum. Separate blenders for the surface and core layers allow different addition rates of resin to be used in the core and surface layers. Different resin formulations or types can also be used with separate face and core blending systems. The mixing for OSB strands is slow and gentle to minimize damage.

Generally particleboard blenders are small compact high speed mechanical mixers – also described as short residence time blenders – with chips remaining within the blender for less than 1 minute (Figure 12.9). The dry furnish enters at one end of a tube containing a central rotating arbor with agitator arms spaced along its length. These throw the particles out to the periphery of the tube while also moving the solid mass along the tube. The resin is injected through nozzles located at the entry end of the tube and projecting through the rotating particle mass around the periphery of the tube. Resin droplets are captured by the particles and further redistributed by the wiping action between particles induced by the movement of the particles within the blender. Fine particles tend to capture more of the resin than do the larger particles, due to their large surface to volume ratios. In some plants, particularly the larger ones, the fines are separated and mixed with a proportion of the resin to ensure that the resin is more evenly distributed between the larger and smaller particles. After blending the fines are mixed with the larger particles before reaching the mat former.

Blender design is essentially empirical, with particular aspects such as blade angle (that determines the direction and rate of material movement along the axis of the blender), rotational speed and nozzle location the basis of performance claims.

'Tack' is important in holding the particleboard mat together as it travels from the forming station to the press. It is difficult to quantify this aspect of resin performance with only subjective assessment of the property. Tack depends on the ability of the liquid resin to form surface tension bonds between adjacent particles in the low density, lightly consolidated mat that leaves the former. It is the result of a complex interaction between the water in the resin, the rate of absorption of this into the dry wood and the size of the resin droplets. The larger the resin droplets the easier it is to form bonds between adjacent particles that are further apart in a lower density mat. While larger resin droplets may contribute to improving the strength of the mat so that it can be transported from the former to the press, there are indications that smaller resin droplets improve bonding efficiency. Too much tack results in material building up on surfaces in the transport and forming areas.

7.4. MDF fibre preparation

With MDF the fibres are much smaller, but have a much greater slenderness ratio than those for particleboard. Fibre dimensions for most species are available, or can be readily determined, so that whole fibres can be characterized accurately. However production of the fibre requires a preliminary breakdown of the wood usually by chipping that will damage some fibres, so that fibre fragments will always be present. The level of these depends on the length of the chip in relation to the length of the fibre. A chip cut to an along-grain length of 25 mm from a material with an average fibre length of 3 mm will cut one fibre in 9 into two parts. Sawdust with a particle length of 3-5 mm will have almost every fibre cut into two parts.

It is these two characteristics – the slenderness ratio, and the overall small size of the particles – that have lead to the distinctive features of the MDF process.

The fibre for MDF is prepared using a low energy thermo-mechanical pulping process. This differs from the thermo-mechanical processes used to manufacture fibre for papermaking in that the thermal treatment is much more intense than for paper fibres, allowing the fibres to be separated while using much less mechanical energy. The thermal and mechanical treatments for chemical pulp, and for thermo-mechanical fibres for both paper and for fibreboard are compared in Figure 12.10.

The chips for MDF are treated thermally to a point where the lignin-rich layer between the fibres softens. This point is characterised by a glass transition point for lignin, generally taken as a temperature of 140^{o}C. To achieve this, the material is heated in a steam atmosphere at a pressure of 8 bar (800 kPa), for a period sufficient to ensure that this condition is reached throughout the material. For chipped material this is achieved in 3-4 minutes. The thermally treated material is then transferred at the prevailing pressure to a refiner, an 'impact mill' that separates the fibres. This sequence in the fibre preparation system, involving the thermal pre-treatment followed by mechanical treatment in the refiner, is shown in Figure 12.11.

The raw material, prepared as chip-sized particles, is delivered to a bin above the preheater, where atmospheric steam is injected to start the heating process. From the bottom of the presteaming bin the chips enter a screw feed that forms a tapered plug

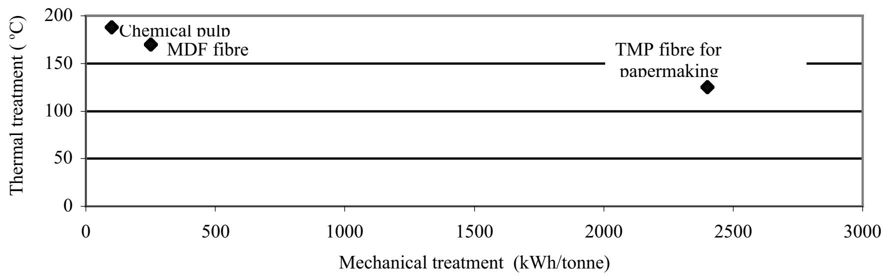

Figure 12.10. Fibre preparation conditions: contrasting primary processing procedures.

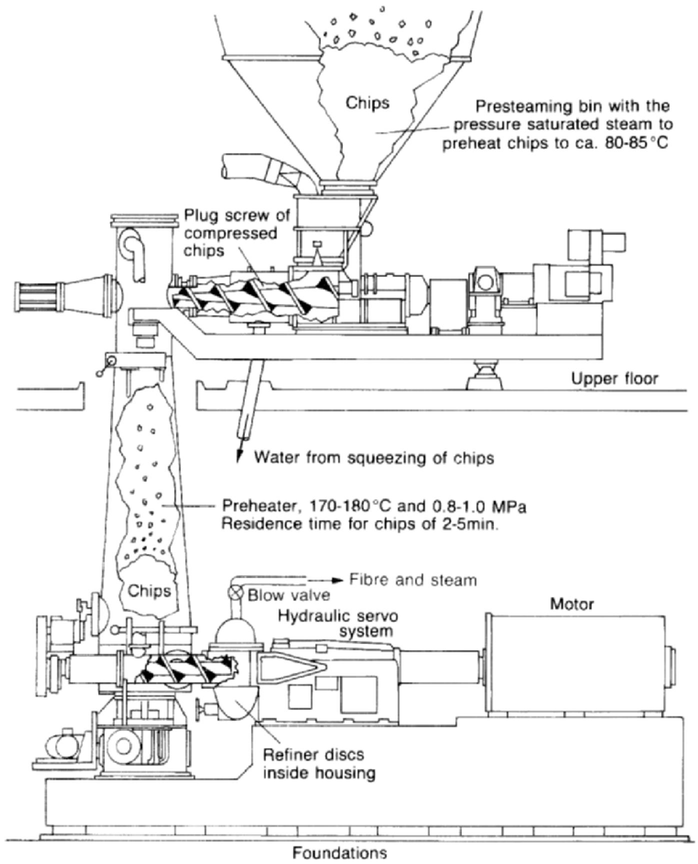

Figure 12.11. Pressurised disc refiner for MDF fibre (courtesy Sunds Defibrator, Sweden).

of material of increased density which creates the seal between the pressure in the preheater and the atmosphere. The speed of the plug screw is controlled to maintain a level of chip in the preheater that is selected to provide the desired treatment time in the preheater to ensure effective fibre separation in the refiner.

The chips move down the preheater pressure vessel and are removed from the bottom by a scraper arm that breaks up the column of material, delivering it to a metering screw conveyor. The speed of this conveyor determines the production rate while delivering the preheated material to the refiner.

Fibre separation is achieved between two discs, one stationary and the other rotating, each faced with a circumferential set of refining plates (Figure 12.12a). The rotating disc has a positioning system, usually hydraulic, that enables the distance between the fixed and rotating disc (0.2-0.4 mm) to be very precisely controlled. Fibre separation is achieved by contact with raised radial bars on the refining plates. The material moves in the grooves between the radial bars but circumferential dams in the refining zone ensure that the material is brought back again into the contact zone between the refiner plates.

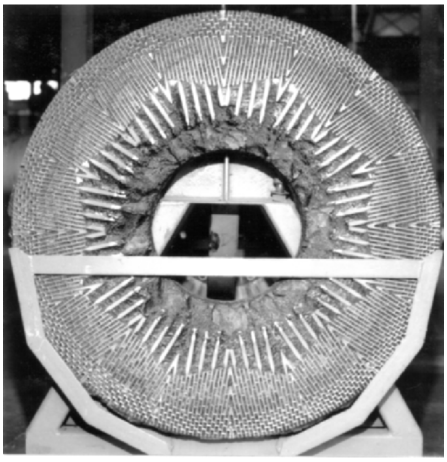
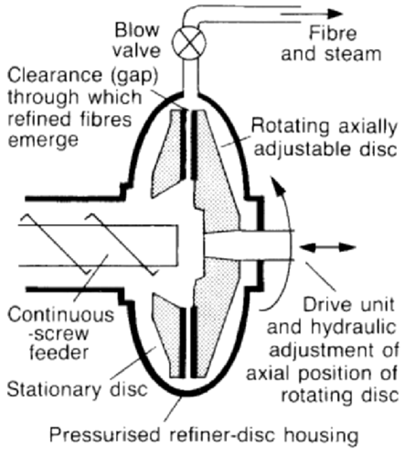

Figure 12.12. (a) Refiner plate. (b) Schematic of the pressurised disc refiner unit.

Refiner plate configuration is the subject of considerable development. Initially the focus was to achieve a satisfactory fibre. Recently this has changed slightly, to minimizing the refining energy consumption necessary to achieve satisfactory fibre quality. Typical refining energy levels are 125-200 kWh/tonne ODF (oven-dry fibre) for softwood chips, with hardwood energies being about 10% lower. New developments in refiner plate design claim to produce acceptable fibre using softwood chip at specific energies below 100 kWh/tonne. Specific energy levels to prepare fibre from sawmill wastes such as sawdust and shavings are higher, with typical refining energy levels of 300-400 kWh/tonne ODF for these materials: no reason for this increased energy requirement has been advanced. Generally the refiner plates used have been developed for chip as feed material and are not optimized for the smaller sawdust particles, or for the lower moisture content of the shavings. Refiner capacities range up to 60 tonne/h for single units with 1.5 m diameter discs driven by motors up to 12 MW in capacity. A typical refiner plate assembly is shown in Figure 12.12b.

7.5. MDF fibre quality

The fibre quality necessary for MDF is a subject of much debate. Reliable assessment of fibre quality is difficult and equipment to measure this is expensive. While panel properties are claimed to depend on fibre quality there are indications that this effect is more likely to be associated with changes in blowline blending conditions than directly attributable to changes in fibre quality. The fibre length for fibre derived from sawdust is probably less than 50% of that for fibre derived from chip. For equivalent strength properties the density may be increased by 5% (725 to 760 kg/m^3) while the resin addition rate may be 10-15% higher than for a panel made with fibre derived from fresh chip. Fibre quality can be mapped as in Figure 12.13, with each quadrant in the length/width axes defining a different area of departure from the ideal case of individual fibres separated without damage.

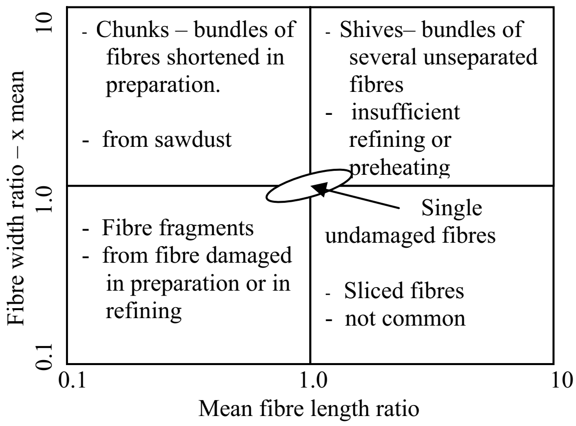

Figure 12.13. Fibre length – width map for defining fibre quality.

There will some fibre components from both the top right and lower left quadrants of Figure 12.13 in all MDF fibre. The shives are only a problem if they are noticeable in the surface of the panel, or if their different swelling characteristics affect overlay performance. Some fibres are always damaged in the preparation of the material before the refiner, so there will always be some fibre fragments. However the operation of the refiner must be such as to minimize these. If the feed material is sawdust or shavings there will be a much greater level of damaged fibres so that the average fibre length will be less.

7.6. Fibre-resin blending

The first MDF was made using particleboard equipment. The fibre was dried and mixed with resin in a particleboard blender. The product had excellent properties, but there were dark areas of varying sizes in the panel that on analysis were shown to have a resin content two or three times the average. These clumps of resin-rich fibre – known as resin spots – were sometimes sufficiently hard as to break saw teeth or to form lumps under overlays. A range of techniques was adopted to break up these lumps, including attrition mills, but with limited success.

The problem was reduced to manageable proportions by changing the point of resin addition to the blowline that carries the fibre from the refiner to the dryer. Resin spots were reduced to an acceptable level, but early experience with blowline resin addition established that higher resin levels were necessary to reach desired property levels. The additional resin requirement was estimated as between 15 and 30%, (Smith, 1998) but there is no published data that allows a direct comparison between the two approaches. One reason advanced for the increased resin consumption with blowline blending is that the thermoset resin has to pass through the blowline and the dryer, both operating at temperatures in the range needed to set the resin. It was claimed (Gran, 1982; Maxwell *et al.,*1984) that this exposure could be sufficient to start the resin cure process.

While dry fibre blending capacity was retained in some plants, the benefits of blowline blending meant that this approach has been adopted for an estimated 98% of MDF produced. Clearly the benefits of the blowline blending approach outweighed the cost of the extra resin needed where this approach was adopted.

Recently dry blending capacity has been installed in some plants with some (typically 20–25%) of the required resin being introduced at the blowline and the rest during the dry blending operation. Resin spotting limits the amount of the resin that can be added at the dry blending stage, but overall cost saving are claimed to justify the cost of the additional blender.

Chapman and Jordan (2003) presented an empirical model of the blowline blending process, based on a fibre-resin droplet interaction model that showed how blowline and resin atomization conditions could be controlled to improve resin efficiency as shown by the IB (internal bond) test. Reductions in resin requirements of 25% are claimed for an optimized blowline, suggesting that the reason for the apparent increase in resin consumption for blowline blending lies in the blowline conditions. The model shows that if the blowline is too large in diameter steam velocities are below those necessary to achieve a good blending outcome in the blowline. If too high a pressure in resin atomizing nozzle is used then the resin jet will hit the far wall of the blowline before atomization is complete. The boundaries for these two conditions limits the resin efficiency that can be achieved.

7.7. The fibre dryer and fibre moisture control

The blowline discharges the fibres, now mixed with resin together with the steam and moisture that accumulated in the refiner, directly into a straight section of the flash tube dryer. These have a straight tube, typically 100 m in length, followed by a cyclone to separate the fibre from the drying air that also conveyed the fibre through the dryer. Temperature measurements along the dryer tube show that most of the temperature drop between inlet and outlet temperatures occurs very shortly after the blowline discharges the fibre and resin into the dryer, typically within 20-30 m downstream from the end of the blowline, i.e. the fibre discharged from the blowline into the drying tube is rapidly dispersed within the dryer airflow. This indicates that the energy exchange between the hot gas and the fibre has been completed within this time. At a gas velocity of 30 m/s the drying time is probably less than one second. Consequently, provided the fibre dispersion condition is maintained, the dryer length can be reduced. In some cases the drying is achieved in a vertical tube rising from the refiner at ground level to the point of the entry to the cyclones that separate the fibre from the drying gas. Entry temperatures range from 100 to 300^{o}C, with exit temperatures usually between 50 and 70^{o}C. The hot gas for conveying and drying the fibre may be heated indirectly with steam or hot oil, but combustion gas, either from dedicated gas burners or from waste combustors is often used, after recovery of heat as steam and or hot oil for other plant requirements, with a consequent reduction in the thermal energy input to the plant.

The exit moisture content from the dryer depends on whether dry blending is used. If the resin is added at the blowline, then the moisture after the dryer is that required in the press, typically 9-13%. If resin is added after the dryer then the moisture leaving the dryer will be lower, so that the water added with the resin will raise the total moisture to that required in the press.

A fibre bin or bunker, sometimes incorporated in the mat former, provides a buffer between the refiner and the press. Bin capacity is quite limited as a tonne of fibre occupies 33 m^3, corresponding to a bulk density for the fibre of around 30 kg/m^3 for a softwood fibre derived from chip. With press lines requiring 10-50 tonnes/h of fibre, the provision of even a few minutes capacity between fibre production and panel manufacture involves a substantial bin volume. There is logic in making this bin as wide as the press line so that the process of distributing the fibre uniformly across the width of the press line can be partially accomplished in the fibre bin.

Press speeds can also be increased if the fibre mat is delivered into the press at an elevated temperature – so reducing the resin curing time. This approach is increasingly being adopted as a means of maximizing press capacity, particularly for thicker panels. The most efficient way of doing this is to maintain as much as possible of the temperature in the fibre as it leaves the dryer. To achieve this, the fibre processing and transport between the dryer and the press must take place under conditions that maintain both the temperature and moisture content of the fibre. The rapid response of the fibre as indicated in the time to complete the drying process requires that both temperature and humidity in processing and transport air must be controlled. The temperature-humidity relationship necessary to maintain a given fibre moisture content in the dryer is shown in Figure 12.14. The figure draws on timber drying data for *Pinus radiata* that is available from a number of sources.

The data applies to all wood particles, but only individual fibres are able to respond in the brief time the material is in a pneumatic transport system.

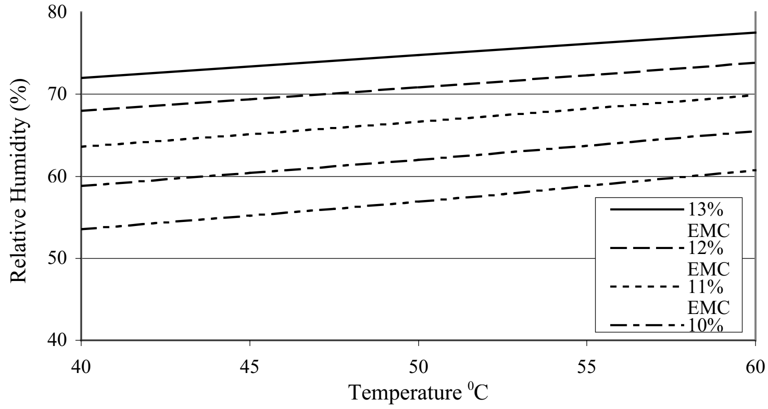

Figure 12.14. Equilibrium moisture content of fibre from the dryer, as function of relative humidity and temperature.

Microwave heaters have been installed on the forming line to preheat the mat (<70°C) prior to pressing. These require space that may not be available in a plant not initially designed for this. Other approaches include injecting steam into the mat, either sequentially from the top and bottom of the mat or in one case, by splitting the mat along its centreline to inject steam before reuniting the two sections of the mat. The mat splitting technique is limited to MDF, while the other approaches can be used with particleboard and OSB mats as well as MDF.

7.8. Mat forming

The major difference between the fibres used to make medium density fibreboard and the particles used to make particleboard is the greater slenderness ratio of the fibres. This is the reason why mechanical blenders designed for particleboard were not satisfactory for MDF. It is also the reason why the bulk density of fibre is low, particularly with fibres prepared from a chipped material. The fibres form networks that are quite strong, with fibres able to span considerable distances. The mats formed from such fibres hold together so that formed or even trimmed edges hold their shape. There is no requirement for tack from the resin to hold the fibre mat together.

There are two approaches to transporting material prepared for the press line. Belt conveyors are used for higher density materials, but the mat becomes very bulky when densities fall. For fibres, pneumatic transport systems, where the fibre is carried by a moving stream of air, are used: although belt conveyors are used for the moving floor of the fibre bin or bunker and for the conveyors that carry the MDF mat from the former to the press.

However pneumatic systems can cause size separation where there is a large range of particle sizes, as with a particleboard furnish. As particle sizes increase so the velocity necessary to carry the material pneumatically becomes greater, so that impact damage also increases. For this reason pneumatic conveying is not used for OSB strands after the screening to separate the strands onto face and core streams. While process considerations determine the type of conveyor selected, belt or chain conveyors have a higher capital cost but lower operating cost (electrical power).

Both approaches are used in the mat forming stage. The object is to form a mat the width of the press, with a weight per unit area such that when compressed to the required thickness in the press, it will have the density desired in the final product, at all points in the panel. Typically the density variation across the width of the mat should be within 2% of the mean value, with density measurements based on 100 mm square samples from the panel. To achieve this it is necessary to distribute the prepared particles uniformly across the width of the pressed mat (the cross direction) and also along the panel (the length direction). The mat former that performs this operation has at times used both mechanical and air transport systems to achieve this; sometimes in the same forming process.

7.8.1. Length direction

To achieve uniformity in the length direction the prepared furnish must be delivered into the mat at a uniform rate sufficient to meet the overall weight of material entering the press. A scale on the forming line weighs the mat as it passes over to provide a control variable that is used to adjust the volume metering rate to achieve the overall weight target for the mat. Two approaches are used. In MDF plants a scalping roll trims the top of the mat, with the material trimmed off being returned to the former. The scalping roll height is adjusted so that the material passing under the roll is that required for the press. The second approach is used for particleboard and for OSB and is being adopted in recent designs for MDF mat formers. In this case the material is discharged from a bin or silo in what is essentially a volume metering device at the rate required by the mat, together with an allowance for side trim.

7.8.2. Cross direction

Uniformity across the width of the mat requires that the prepared particles be distributed uniformly across the width of the mat. This can be measured off line as discussed above, but it is usual to provide online measurement of weight variability across the mat by a scanning density gauge that measures the density of the mat by absorption of radiation. These instruments indicate persistent areas where the weight per unit area is outside the desired limit. To achieve the weight profile across the mat requires some means of taking the uniform mass flow rate determined by the overall mat weight requirement and spreading this evenly across the width of the mat. With MDF mat formers this has been achieved by displacing the airstream carrying the fibre to the mat former from side to side. Both mechanically driven oscillating nozzles and air jets have been used to achieve this. To adjust the placement of MDF fibre across the width of the mat the nozzle mechanism can be adjusted, or a variable vacuum applied beneath the scalper roll to change the density of the mat at specific points across the width of the mat.

Recent designs for fibre mat formers have tended to move towards the mechanical formers used for particleboard and for OSB. As noted earlier, the low bulk density of prepared material requires large storage volumes. If the storage bin (or bunker) is made the same width as the forming line then it becomes part of the mat forming system. In this case the material is distributed across the width of the bin by an oscillating conveyor, moving across the width of the bin. By controlling the speed of the oscillating movement at each point as the conveyor moves across the width of the forming line the weight distribution across the mat can be controlled.

7.8.3. Orientation

OSB strands are oriented, along the mat on both surfaces, and across the mat for the core or central layer of the mat (Figure 12.15). To form the face layer a disk screen is used to align the strands lengthwise along the mat. Once aligned the strands fall through between the discs onto the mat. The disks align the strands until they are able to fall between the disks. By increasing the distance between the rotating shafts along the length of the former it is possible to achieve some sorting by length of the flakes.

For the core layer of the OSB mat the flakes must be able to fall free. They are caught in a series of rolls running across the width of the mat, with radial pockets that the strands fall into. The pockets then discharge the aligned flakes so that they fall across the width of the mat. There are three forming stations for OSB. First the bottom face is laid on the moving forming belt which then moves through the cross forming station that places the core flakes. Finally the top face of the mat is laid down to complete the process.

Particles falling onto a moving belt develop an inherent degree of alignment arising from the inertial forces involved on first contact with the mat. This alignment in reflected in a higher bending strength in the length direction of the mat than across the mat, a feature common to both particleboard and MDF. The edges of the mat are influenced to some extent by the board against which the edge is formed and it is usual to trim this after forming, returning this material to the forming station.

7.8.4. Particleboard mat forming

Particleboard formers use several techniques to achieve a smooth transition between the smallest and the largest particles in the mix. To achieve a symmetrical particle distribution from top to bottom of the panel the forming stations also need to be symmetrical, so that the top face of the panel is laid over the bottom face using a mirror image of that used for the bottom face. For the surface layers of the mat a wind sifting former is often used. This uses a controlled flow of air to separate the particles, the larger heavier ones dropping more quickly, the finer ones being carried further before dropping out of the air stream (Figure 12.16). The degree of separation is determined by the air velocity. The wind sifting former can be

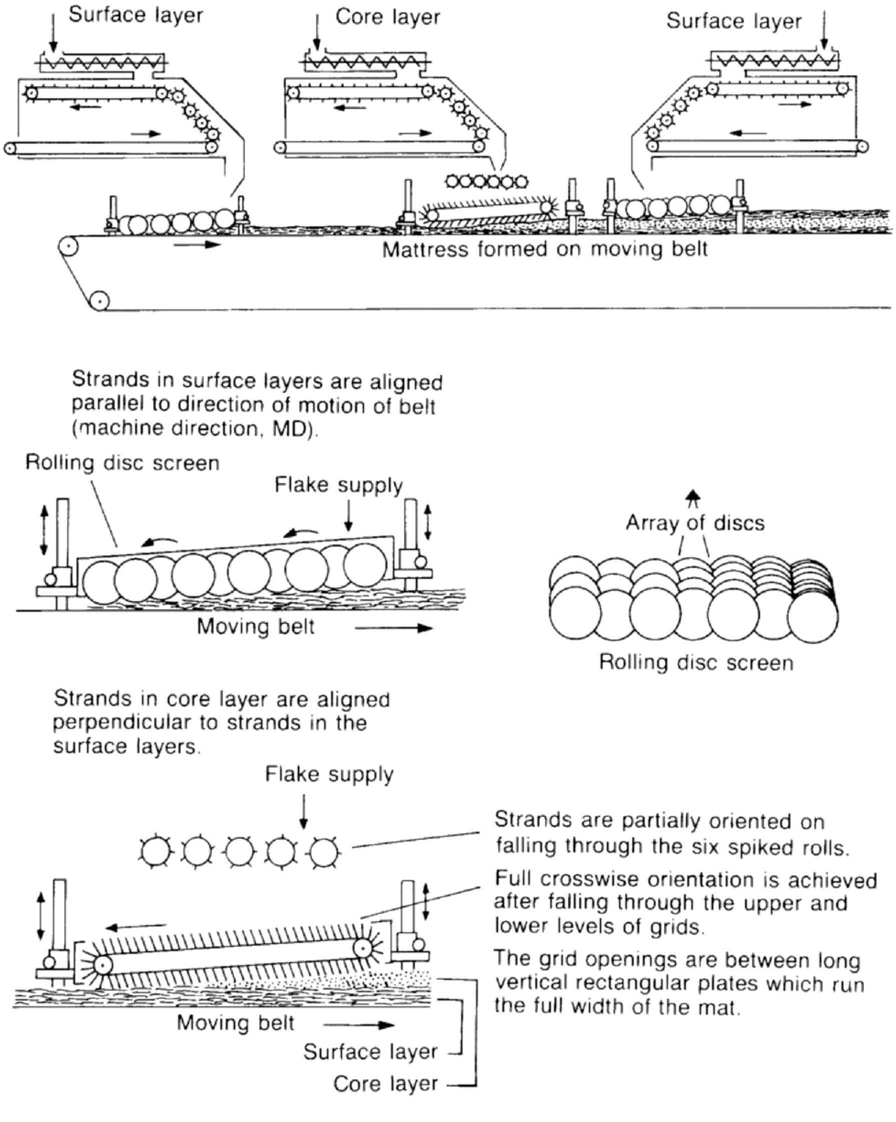

Figure 12.15. Orienting of strands along and across the mat for OSB. (courtesy Siemplekamp Krefeld, Germany).

configured to place the finest particles at the outer layers of the mat, or reversed so that the fines are placed in the centre of the mat thickness. Alternative forming arrangements use variations on the disc screen concept, with rotating elements agitating and aligning the particles while at the same time presenting an opening that particles of a given size are able to pass through. By varying the roll spacing and configuration a graduation of particle size is achieved. The mechanical formers are claimed to have lower power requirements, and do not have particulate-carrying air streams that may require cleaning.

7.9. The mat forming line.

The forming line transports the formed mats to the press. It is a conveyor system that may incorporate specific features to:

- Improve mat properties by cold pressing to increase its density. This increases its strength and makes it thinner so that the mat requires a smaller opening in order to enter the press.
- Apply moisture or other materials to the surfaces of the mat to increase initial heat transfer rates in the press or to modify mat characteristics in some other way: this includes release coatings to reduce sticking to the press platens or the steel bands of a continuous press.
- Trim the sides of the mat to remove the outer portions where the mat characteristics are affected by forming or precompressing operations.
- Cut the mat to the required length for batch presses.
- Monitor mat properties for overall mat weight, variability across the mat width and for the presence of metal or other material that could damage the press.
- Allow mats that contain foreign materials, or with weight variability outside density specifications to be removed before the press.

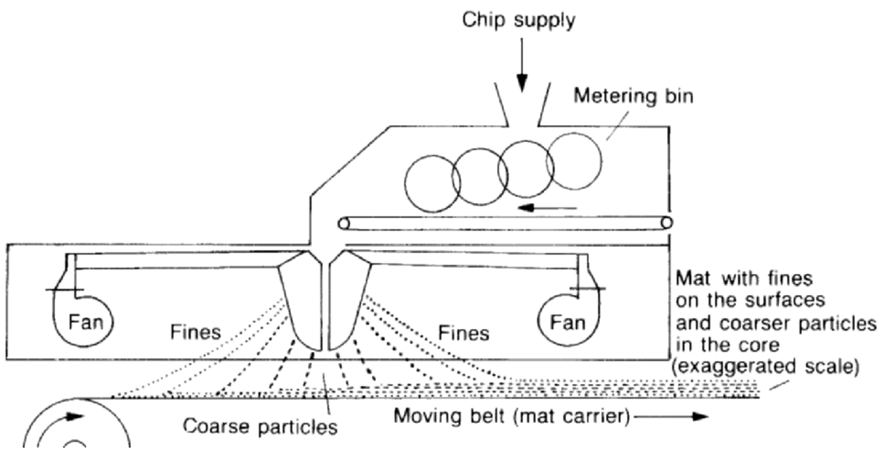

Figure 12.16. Schematic representation of an air sifting forming head producing a graded particle size distribution. The manifolds are alternately forward and backward facing vertical slits offset to allow the particles to be carried through by the airflow (courtesy Bison Werke, Springer, Germany).

7.9.1. The precompressor

The precompressor is a cold press that compacts the mat. Precompressors are used on almost all MDF lines, and also on some particleboard lines. The objective is to reduce the thickness of the mat from the former by increasing its density. As this occurs much of the air in the mat is expelled, reducing the effect of this on the compression of the mat and on heat transfer in the hot press. The mat becomes stronger as a result and is better able to traverse gaps between conveyors on the forming line, and pass through the side trim saws without damage. It also reduces the size of the press opening necessary to accommodate a mat of given weight. For an MDF mat, the formed density of the mat is very low (*c.* 30-50 kg/m^3), resulting in a very fragile mat. Batch precompressors have been used on batch press lines, following the example of the pre-presses used in plywood manufacture. However nowadays continuous inclined belt precompressors are standard.

As the mat travels down the line an inclined top belt makes contact, slowly compressing the mat as it moves through the press. The inclined belt is normally woven to allow air displaced from the mat to escape through the belt. As the force required to compress the mat increases the loading rolls become larger, and high strength belts are needed to move the mat through the zone of maximum compression. The precompressor is able to compress the mat to the density of *ex*-press panel. The thickness and density changes through the precompressor for an MDF mat are shown in Figure 12.17. The height of the MDF mat is reduced from 20-25 times *ex*-press panel thickness to 8-10 times this for a softwood fibre, measured after both the elastic and viscoelastic recovery of the fibre. Most of this reduction in height is due to a rearrangement of the fibres within the network laid down in the mat former. A study (Thorbjornsson, 1985) showed that the reduction in mat height was greater where a force was applied and released several times than where a greater force was applied in a single application. This suggests that the reduction in load allows the fibres to move relative to one another so that the lower mat height is achieved without the residual strain that arises where fibres are unable to move within the fibre network.

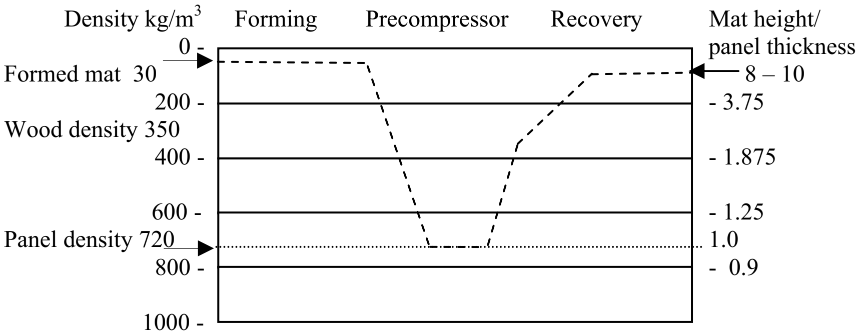

Figure 12.17. Density changes and mat height (as a multiple of the desired panel thickness) in a precompressor on an MDF forming line.

The stronger mat after the precompressor is better able to cope with the edge trimming saw and, in the case of a batch press, the cross saw that divides the mats into the length necessary for the press. It also enables the mat to bridge the space between the conveyors necessary to carry the mat into the hot press. One of these conveyors is able to retract to create an opening in the forming line that allows a mat containing foreign material to be removed ahead of the press.

The mats for particleboard are of higher initial density than those for MDF, typically 150-200 kg/m^3 on forming, so that a precompressor is not required to reduce mat height. However the particleboard mat tends to be fragile as a result of the reduced slenderness ratio of the particles and relies on 'tack' between the particles to maintain mat integrity. The precompressor is needed to improve the mat strength so that gaps between conveyors do not cause mat damage. As will be seen in the discussion of press types changes between conveyors are necessary in multidaylight presses and with continuous presses where the mat must be transferred between the forming belt and the steel belt that conveys the mat through the press. Precompressors are not used on OSB lines.

The forming line may also include the means for heating the mat ahead of the press. Microwave heaters have been used. Other systems apply steam or a steam/hot air mixture to heat the mat. Typically the mat can be heated to 50-60^{o}C before the press. Temperatures in excess of this are likely to result in some resin cure ahead of the press.

7.10. The hot press

In the press, the prepared material is subjected to pressure to compress the material to the required thickness and to heat to cure the resin to form the bonds that give the panel the required strength and other properties. The press has a major impact on the plant design.

- The platen size determines the maximum size of the panels that can be produced and how these can be cut into the sizes required by specific markets. The press size can easily be specified for plants serving a local market with a preferred panel size. The decision is much more complex for plants seeking to meet the requirements of diverse international markets.
- The total area of the platen(s) determines the capacity of the plant. Once installed it is not easy to increase this, so that this limit needs careful consideration when equipment is sized.
- The rate of press closing, particularly under the high pressures that develop in the initial compression of the mat, has an important impact on the development of the density profile through the thickness of the panel. For a batch press the hydraulic pump capacity determines the closing rate. Increasing this rate reduces the quantity of material in which the resin has cured before the required face density has been achieved, and which must be removed in sanding.

7.10.1. The batch press

Initially particleboard was produced in batch presses developed from those used in the plywood industry. New approaches to transferring the fragile particleboard mats into the press were necessary. These required a greater distance between the platens to allow the thicker mat on its carrying tray to be loaded in the press, so reducing the number of openings that could be accommodated within a given press structure. On the other hand the particleboard mat was not subject to the size limitation of veneers, so that the platen could be both wider and longer than those used for plywood production. The benefits of larger ex-press panels were clear both in terms of minimising edge trim loss and in flexibility of cutting into panel sizes required by the market. Platen widths of 2.4 m are common in batch presses for particleboard and MDF, with continuous presses up to 3 m wide for MDF. Wider presses are used for OSB, with the widest press able to produce 3.6 m wide panels (Natus, 1996).

Batch presses are configured either as multidaylight, or single opening systems. The multidaylight press has a number of platens stacked vertically in a structural frame. The intermediate platens are held in position by a simultaneous closing system that ensures that the overall movement of the press is distributed evenly between the platens. This system of levers ensures that the weight of the platens above a particular opening in the press does not affect the loading. It is not strong enough to overcome the compressive load of the press, and incorporates springs or other mechanisms to allow the platen to move in response to the resistance of the mats on either side of the particular platen as loads increase.

The multidaylight press system requires a loading structure that accumulates the mats for the next press load while those in the press are being compressed. The loader has positions for each mat spaced according to the distance between the platens when the press is fully open. The loader moves vertically so that mats from the press line can be loaded into each position. When all mats for the next load are in position and the press operation is complete, the press opens. The loader trays carrying the mats move into the press, pushing the pressed panels out into an unloader that accepts the panels from all openings. As the loading trays retreat conveyors in the trays deposit the new mats directly onto the platen surfaces. When the loader trays are clear of the press it can close. Loading the press takes a fixed time, usually somewhere between 30 and 50 seconds.

The other type of batch press is the single opening type. As the name suggests this press has only one opening, but uses very large platens – up to 2.4 m wide and up to 27 m long. This system uses a steel conveyor belt that runs from the forming station through the press. The mat is formed on this conveyor while the panel is in the press. As the conveyor belt is stationary while the press is closed the mat former moves along the belt to lay the mat down. When the press operation is complete the press opens and the conveyor belt moves to discharge the pressed panel while moving the prepared mat into the press. With the mat travelling on a single conveyor mat integrity is less of an issue. The press is a downward acting one with the hydraulic cylinders acting to push the top platen down on the mat. This allows the press to open just sufficiently to allow the mat to enter the space between the platens, so that the press closing time is minimized.

A further variant of the batch press is the steam injected press. The platens have holes that allow steam to pass from headers through the platen into the material in the press, where it condenses to heat the material directly. When the resin has cured the headers can be connected to a vacuum system that reduces the pressure within the mat, so that steam flashes off and is drawn back out of the panel. The additional complication of a steam supply means that this approach has been implemented only in a single opening press. The approach is useful for very thick panels. Here press times would otherwise be very long because the heat must transfer from the platens to the centreline of the mat, but there are some limitations. While the steam travels rapidly through the mat, often with the flow from one side being delayed slightly to avoid the air in the mat accumulating in the centre, it has to pass through the faces of the panel. This limits the face density that can be achieved in the steam press.

7.10.2. Continuous presses

A continuous press was long an objective in the industry. There are a number of benefits of continuous pressing. From the panel perspective, the ability to determine the length of the panel after the press provides greater flexibility in panel size. Trim loss is also reduced.

Continuous presses also have no dead time associated with loading the mat and unloading the panel after the press, and are thus able to manufacture thin panel more economically than batch presses. There are at least four manufacturers of continuous presses, and today this type of press would normally be selected for a particleboard or MDF plant.

The first fully commercial continuous press was the Bison Mende calendar press (Figure 12.18.). This adopted the large diameter heated calendar rolls used in the paper industry to dry tissue papers. The heated calendar roll provided one ‘platen’. The mat was formed on a continuous steel belt that carried the mat to the calendar roll and then around the roll. Additional heated rolls applied pressure and heat to the mat at points where maximum pressure was needed. The resin cured as the mat was heated and the finished panel left the calendar roll as a continuous strip that was taken to a sawing station. The steel belt returned to the forming station. In the Mende press the panel is formed and the resin cured in contact with a curved surface. The panel is straightened by inducing a reverse curve in the panel as it leaves the press, but this limits the thickness of the panel that can be produced. Calendar diameters of 3, 4 and 5 m are used. It is claimed that 12 mm thick panel can be produced on the 5 m diameter calendar roll. In practice this type of press is rarely used to manufacture panel above 6 mm in thickness.

The continuous production of thicker panels needed a straight line path through the press that required new solutions to moving the prepared mat through the heating and pressure zones needed to create the panel (Figure 12.19).

The first straight through press was the Küster press (now owned by Metso), this was followed by presses using a similar design by Siemplekamp and Dieffenbacher, both major suppliers of batch presses. The straight through press design uses two

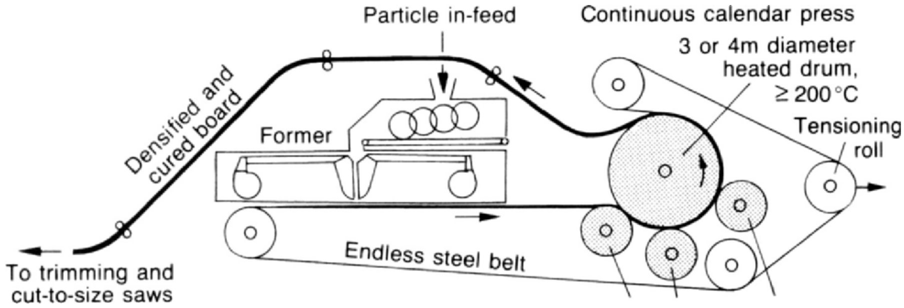

Figure 12.18. The Bison-Mende thin panel press cures the panel as it moves around a large diameter heated calendar roll (courtesy Bison Werke Springe, Germany).

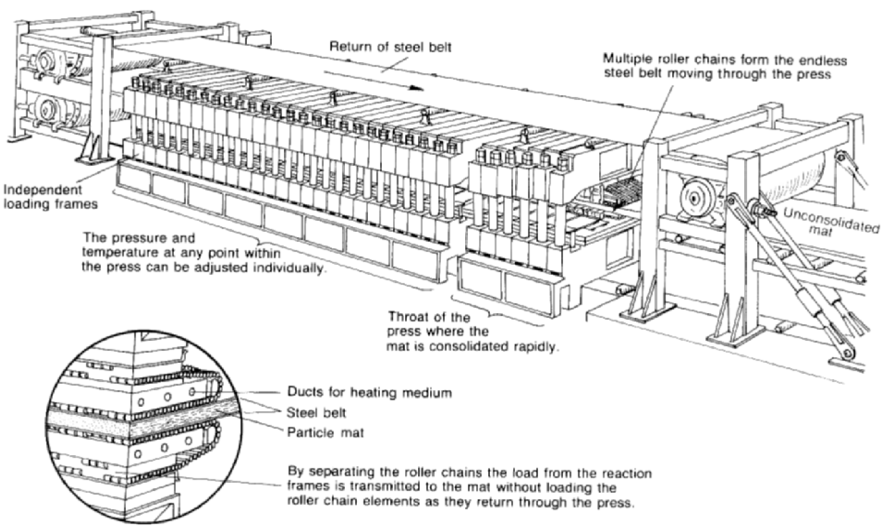

Figure 12.19. The Küsters continuous press. Similar presses are able to press a panel up to 3.1 m wide, with a press length up to 60 m. Maximum speed of the panel through the press can above 60 m/min. (courtesy Küsters, Krefeld, Germany).

steel bands to carry the material through the pressure and heating zones of the press. Some means is needed of controlling the friction between the moving steel bands and the stationary platen. In the case of the three suppliers just mentioned, a system of rollers between the steel band and the heated platens operates as a roller bearing to minimize the friction, while also transferring heat from the platen to the steel belt and then to the mat. Another straight through design by Bison uses a circulating oil film between the platen and the steel bands to control friction and transfer heat. This has not been particularly successful in the market.

There are other advantages in the continuous press that are specific to pressing panels.

- The press loading time is eliminated, with the mat compression starting as soon as the mat enters the press.
- Press design can be optimized, with the different stages of the pressing process occurring at different points along the press; the structure and configuration of the press can be adjusted to suit the pressure and temperatures required at that point in the press cycle. High pressures are only needed for the initial compression of the mat, perhaps 25% of the total press length. This level of pressure is not required in the middle part of the press so that the design pressure can be reduced. It is usual to provide a high pressure zone at the end of the press to allow the final thickness to be achieved. Platen temperatures can be varied along the length of the press. In one design, the last 25% of the press length can be configured to cool the mat, reducing the steam pressure within the mat.
- The platen distance can be set at each frame of the press so that a reproducible time – platen opening profile can be achieved.

Continuous presses have become the preferred approach for MDF and particleboard plants. The capital cost is higher than for an equivalent capacity batch press line but in most cases the benefits in terms of reduced material losses and improved product quality outweigh this. Continuous presses are also used for OSB production and for LVL manufacture, a product where the ability to produce material in a range of lengths is important in meeting market requirements.

7.10.3. Pressing time

The panel needs to be in the press long enough for the heat from the platens to reach the centreline of the panel and cure the resin, forming bonds of sufficient strength to allow the panel to be handled as it leaves the press. This depends not only on the platen temperature (or that of the steel transport bands in the case of a continuous press), but also on the rate of heat transfer through the mat. This is influenced by a number of factors such as the density of the panel and the resin type, and most significantly on the moisture content of the material in the press.

An empirical method is used to determine the pressing time and so the press size necessary to meet a particular capacity. Typical press factors for continuous presses are given in Figure 12.20 for particleboard, MDF and OSB.

Press factors are quoted as seconds/mm panel thickness. Generally press factors increase with thickness as the distance the heat has to move through the panel increases. But the curves are not smooth. The discontinuities result from changes in panel density or pressing strategy. Press factors are lower for continuous presses, a result that might not appear logical considering the need to transfer heat from the hot

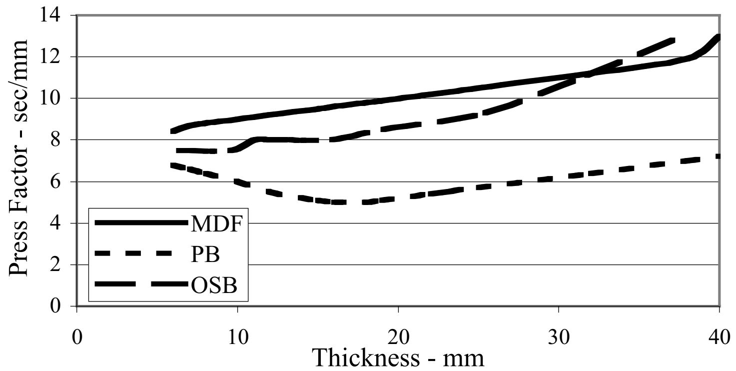

Figure 12.20. Typical continuous press factors for particleboard, MDF and OSB of various panel thicknesses using a continuous press (courtesy, Dieffenbacher, Eppingen, Germany).

platen, through the rollers and into the steel bands before it can move into the mat. However the rapid initial close of the press, together with the level of control of temperature and platen distance along the press result in reduced press times.

Press factors are used to determine the relationship between press size and plant capacity in the design phase. For an operational press, the press factors become a reference point with operational strategies generally resulting in press times below those determined by the press factors.

The press capacity (m^3/h) for panel of a given thickness is derived as follows:

$$\text{Press capacity} = 3.6\ n\ L\ W\ t/(t\ p_t + K)$$

where n is the number of daylights (n = 1 for a continuous press), L and W are the press length and width (m), t is the panel thickness (mm), p_t is the press factor (s/mm) for this particular press type/panel type, and K is the loading time (s) for a batch presses.

7.10.4. Press control

The objective in closing the press is to compress the material in the press to the required density while heating the material so that the resin cure will occur more rapidly. The press has the capability of compressing the material to a very high density, well beyond that required for the product, and must be controlled so that the desired final density of the panel, determined by the final thickness of the panel, is achieved.

With a plywood press, there is very little advantage in compressing the material beyond that point necessary to achieve a satisfactory level of bonding. In fact with the bending strength of the panel proportional to the square of the thickness: any reduction in thickness will adversely affect the bending strength of the panel.

However the property levels of both particleboard and MDF are density dependent, and this is used as one of the controllable variables for these properties.

Initially the objective was simply to control the closing of the press to the point where the desired panel thickness was achieved. The next objective was to reduce the closing time to maximize the press capacity. This required a more sophisticated approach to control of the press hydraulics. Distance sensors to measure the platen distance provided new opportunities for improved control. For a single opening press, position sensors allowed control of panel thickness at each sensor.

With multidaylight presses the distance sensors measured the aggregate distance over all openings. The thickness in individual openings was not controlled, and depended on relative mat conditions in adjacent openings, as well as the straightness of the platens at that point in the press: platen bending is not uncommon, particularly if a mat is incorrectly loaded in the press. To overcome the weight of the platens a system of simultaneous closing arms, designed to ensure that the weight of the platens does not contribute additional load on the mats below, is used. At first, these did not have the capability to impose additional loads on individual mats, but as multidaylight presses have responded to the improved control possible in a continuous press, thickness control on individual openings through a simultaneous closing system designed for greater loads has been offered by some manufacturers.

The straight through continuous presses enable very precise control of the process to be achieved. As the mat moves through the press, the time axis of the press cycle corresponds with the distance along the press, so that the different stages of the pressing process occur at different points along the length of the press. This allows the design of the press to be optimized for the particular process that is occurring at that point in the press. The platen temperature can be varied at different points along the press with one design, the Küster press, now offered with a cooling zone in the latter part of the press.

A number of studies have examined the relationship between press closing and panel properties, specifically bending strength. Overall, the average density of the panel is determined by the mass of material in the press as it closes and the thickness of the finished panel. However the distribution of material within the thickness of the panel can be controlled to give the desired density profile through the thickness of the panel. This profile, known as the vertical density profile (VDP), has a significant effect on the properties of the panel. Typical vertical density profiles for particleboard, MDF and for OSB are shown in Figure 12.21.

The vertical density profile provides a means of optimizing the press operation, as well as panel performance. While initially undertaken in the laboratory, the VDP can now be measured on the moving panel as it leaves the press, using an x-ray backscatter technique (Dueholm, 1996). This is an expensive tool, but the purchase is increasingly common as the value of real time VPD information is appreciated.

The density profiles reveal density areas at both surfaces, with a lower density in the centre of the panel. The high surface density gives good bending strength and surface finishing characteristics. A low density in the core of the panel allows these surface properties to be achieved with an average panel density that is only 60-70% of that at the surface. For particleboard and OSB the profile between the peaks is a

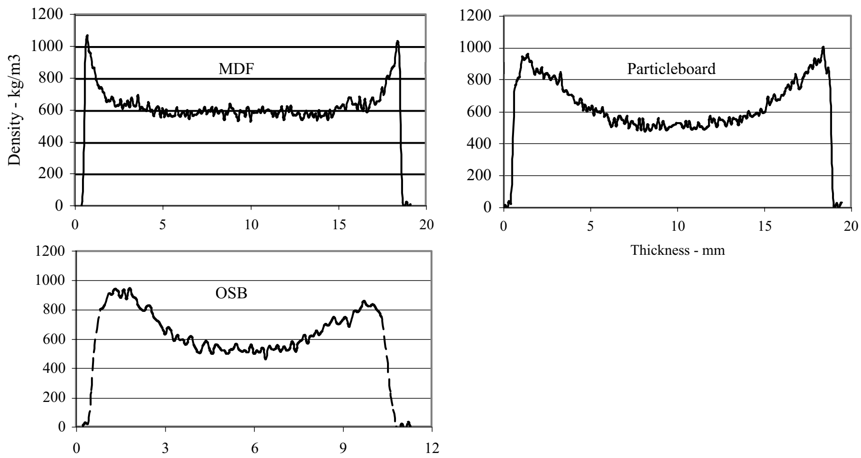

Figure 12.21. Vertical density profiles for particleboard, MDF and OSB.

relatively smooth curve. For MDF the density peak is more sharply defined, with a definite transition between this and the flat density in the core part of the profile.

The density profile results from differences in the compressive response of layers within the mat. Closing the press creates a strain on the mat; the mat responds to this strain depending on the local stress–strain relationship at each point. Apart from density, the only variables that change (both in time and position) during the closing of the press are the moisture and temperature within the mat (Figure 12.22). The degree to which gradients in these variables are established during the press closing determines the density differences between face and core that can be generated. The difference between the MDF profile and those for particleboard and OSB suggests that such gradients are more easily established within the fibres of the MDF mat. Preheating the mat ahead of the press also reduces the ability to create such gradients within the mat, making it more difficult to generate density profiles with sharply defined features. This situation has been modelled (Figure 12.21) with the objective of predicting the VDP of the panel from the press closing conditions, the initial conditions of the mat as it enters the press and the temperature and moisture changes within the mat as the press closes.

The VDP provides a means of identifying the contribution of the press in determining panel properties, separating this effect from other possible causes such as resin mixing and performance, or particle size/shape effects. It also recognises the significant effect of density on the properties of panel products.

7.10.5. Processing panels after the press

After the panel leaves the press it is cut to the required length (for continuous presses) and the edges trimmed. Panel thickness is measured at several points across

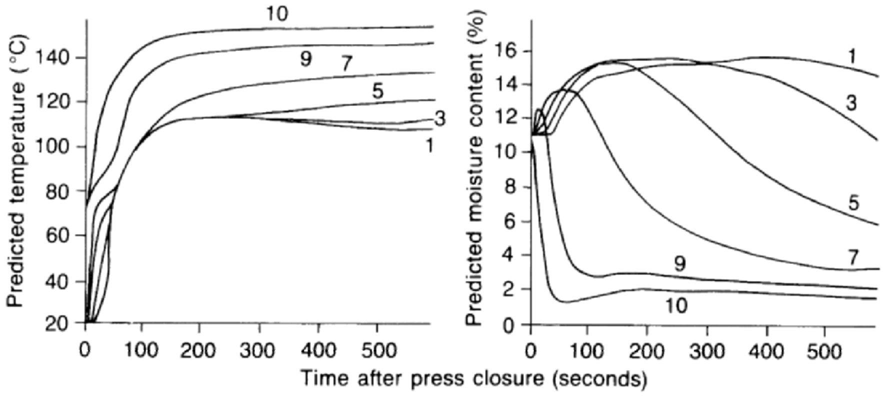

Figure 12.22. The effects of time on temperature and moisture content within a conventionally pressed panel are shown for different layers within the panel. Layer 1 spans the midpoint of the thickness of the mat while layer 10 is in contact with the hot platen (Humphrey and Bolton, 1989).

the width of the panel and acoustic transmission through the panel used to identify areas where delamination within the thickness of the panel has occurred. Panels bonded with UF resins must be cooled before stacking. If the temperature in the stack is above 65-70°C then hydrolysis of the resin can occur resulting in a loss of bond strength. Cooling is achieved by holding the panels in individual slots in a cooling wheel that allows each panel to lose heat to the surrounding atmosphere. During the cooling period the temperature of the panel falls until the temperature in the subsequent stack is below 60°C. During this time the moisture profile through the panel also equalises, with moisture moving from the centre of the panel back to the faces. If the cooling process is too rapid the *ex*-press moisture profile will be frozen in the panel leading to subsequent instability. Cooling times are usually based on 30-45 min in the cooler for the thickest panel produced on the press line. MDF and particleboard panels are then sanded to the finished thickness, with a finishing grit ranging from 120-180 to provide the necessary surface smoothness.

OSB is generally not sanded to a smooth surface, but one face is deliberately roughened, either by a patterned platen in the press, or by a coarse sanding, to provide a non-slip surface.

8. PRODUCT STANDARDS AND PANEL PERFORMANCE

Product standards provide established performance levels that have been found satisfactory for typical end-use applications for a given product. Standards evolve in that they build on methods and approaches that have been found to be suitable for similar products. Initially standards were local product specifications. These have been extended both in range and in scope firstly as national standards and increasingly as international standards to cover products in international trade. Often product standards are specified as a means of compliance with building codes.

For panel products these standards provide both test methods, together with the property levels required to meet specific performance levels. Product standards can be either be prescriptive, specifying the permitted means necessary to achieve a particular end, or performance based, where a particular property level must be reached but the means for achieving this are left to the manufacturer. Most standards are a mixture of both. Density is not generally specified, apart from a general range: MDF is specified in some standards as lying in the range from 400 to 900 kg/m^3.

Standards may cover:

- Tolerances on panel dimensions – length and width, squareness and thickness.
- Strength properties – bending strength, bond strength.
- Environmental issues – formaldehyde emission characteristics.
- Response to water.
- Durability assessment.

The strength properties required are generally the bending properties, measured as Modulus of Rupture (MOR) and less commonly as Modulus of Elasticity (MOE). Both these are significantly influenced by the surface density of the panel. The tensile strength across the thickness of the panel (also called internal bond or IB) is extensively used to measure the bond strength. In this test a sample from the panel (typically 50 mm square) has loading blocks glued to each face of the panel. When the glue has set a load is applied across the thickness of the panel until failure occurs. The calculated stress at failure is the IB value. The test is relatively simple, but has a high coefficient of variation. Failure generally occurs at the lowest density point in the thickness of the panel so that the IB correlates best with the minimum density point on the profile. Even so, correlation coefficients between IB and minimum core density for MDF are only in the range 0.6-0.7. Despite its limitations, the test is used extensively, and is the main test used to monitor the resin level in particleboard and MDF; this is adjusted to maintain the IB at the required level.

8.1. Formaldehyde emissions

Formaldehyde emissions are a significant factor in panel performance made with formaldehyde based resins. The formaldehyde is released from the panel at a rate that reduces over time, but this can be sufficient to raise formaldehyde to levels regarded as hazardous, particularly in living spaces. Various methods are used to assess the formaldehyde emission characteristics of a panel, ranging from the toluene extraction method used in Europe (the perforator test) to a number of chamber tests that measure the formaldehyde emitted from panel samples in an enclosed vessel. There are a range of rating classes, the most commonly used are the E ratings, where E3 is the highest emission class rating (rarely used today) through E2 to E1 levels, the most commonly specified level. E0 is a very low emission panel, while a new class of super E0 has a formaldehyde emission level close to that of wood itself. Formaldehyde levels in occupied spaces are generally controlled by regulation, with the use of a panel with a specified formaldehyde emission characteristic accepted as a means of complying. The development of formaldehyde – based resins able to meet the decreasing emission levels has been a major focus of UF resin development over the past few years. Initially UF resins with low emission characteristics required higher resin addition rates to maintain strength properties and proved to be more sensitive to plant and furnish conditions. Further development has recovered some of these disadvantages. Formaldehyde emission is limited to UF and MUF resins. PF, although a formaldehyde resin, is more stable and is classified as a low emission system. Isocyanate resin systems have no formaldehyde emission from the resin and can be used in situations where formaldehyde cannot be present at any level.

8.2. Response to water

A range of methods evaluating the response to water, either as liquid or as humid air are used in standards. The methods involve a short term response to immersion to water at a controlled temperature for a specified time with measurement of the resulting change in panel dimensions, particularly perpendicular to the plane of the panel. The swelling in this direction is a reversal of the compression locked by the formation of the adhesive bonds in the press. This response is density related and can lead to anomalous results if the test regime does not provide sufficient time for the water to completely penetrate the sample. Increasing density slows down the rate at which water travels through the sample, but because the compression is greater, a greater degree of swelling will result. Increasing density can give a lower thickness swell result in a short term test, but the long term swelling may in fact be greater. For products designed to operate in interior situations where exposure to moisture is not anticipated, resistance to water is not a major issue and the water response tests indicate whether wax added as a sizing agent is present.

For products intended to perform where a degree of water exposure is likely, such as kitchens or bathrooms a higher level of resistance to moisture is expected. In this case the permitted swelling level is lower, and additional tests may be required.

8.3. Durability

The response to water is critical in applications where water exposure occurs. Both particleboard and MDF ultimately fail unless moisture is excluded, limiting applications to those areas where the chance of this occurring is small. Most applications in furniture, fittings and cabinetry meet this requirement. There are also applications where structural performance is not usually critical, that is where a failure of the item does not endanger the structure as a whole.

Where a component contributes to structural performance in a building then a durability requirement, usually expressed in terms of exposure conditions and time, will apply unless the possibility of moisture affecting that part of the structure can be totally excluded. For established materials the durability performance is known.

For newer materials, particularly the adhesive bonded composite panels being considered, prediction of long term performance is much more difficult issue. There is an established history that suggests that some durability requirements can be met. OSB is used as a structural element in buildings, either as a bracing panel or as a component of a beam. Much of the time between the development of panels based on wafers or strands and the commercialisation of OSB was used to establish the durability of the material. Particleboard is used as a structural flooring panel in both Australia and New Zealand. The product formulations used are very different in the two countries, possibly reflecting the differing exposure conditions. In New Zealand a UF bonded particleboard has been used for some 50 years. It has proved to be satisfactory, even when exposed over a 2-3 month construction period. The Australian requirement is for a panel with a given MOR retention after a boil test, requiring the use of a PF adhesive, or one with similar response in a boil test.

There is a range of accelerated ageing techniques available that simulate the effect of moisture change in particular as well as ultraviolet radiation. For panel products it is the moisture change effect that is most significant. In order to simulate the effect of long term exposure, most accelerated tests use shortened time scales along with stress levels that are greater than those encountered in actual service conditions. There is a continuum between the stresses encountered in service conditions and the increasing levels of stress used in the accelerated tests and those used as quality control tests as shown in Figure 12.23.

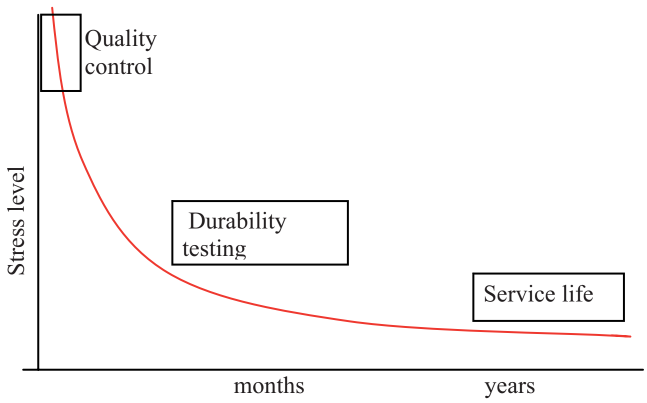

Figure 12.23. Stress-time relationship for durability assessment.

The extreme case is that induced by placing the material in boiling water. This test was originally used to distinguish between PF and other resin types. It has since been used in conjunction with IB or MOR tests on the sample after boiling to measure strength loss. The boil test generates a severe stress on the panel, one that has the advantage of being predictable, but the relevance of this test to actual exposure conditions has been questioned (Carroll, 1980). It has been used as a ranking test (Lehman, 1977, 1986) to screen a large number of samples quickly. Other tests use several cycles of stress, usually involving immersion or exposure to water, followed by drying and in some cases freezing. This test appears as the cyclic test (V313) in many standards (NZS, 4266.11) or in a similar test (ASTM, D1037-72a). In the cyclic tests the stress level is reduced, but is applied in several cycles (3 in European cyclic test, 6 in the US). There are a number of proprietary accelerated weathering tests that simulate selected climatic conditions in a shortened time period. The effect of the accelerated weathering on wood–based panels is evaluated using standard strength tests.

Fibreboards have been used as exterior sidings, a non-structural application, but with limited success. These examples indicate that while some applications where durability is a requirement are successful, these require control of the exposure conditions, particularly those that can lead to moisture ingress.

The durability assessment may also include measurement of the response to insect and fungal attack. There would appear to be little inherent resistance to these forms of attack beyond the characteristics of the wood. A range of insecticides and fungicides can be added. This is easiest if these can be mixed with the adhesive system provided that resin performance is not compromised. The size of the particles means that the selected material can be distributed more easily within the panel. However insect/fungal attack usually does not become a factor unless the moisture content increases, and it is this response that is most important.

8.4. Factors influencing panel durability

Selection of resin type is significant in the performance of wood–based panels. UF resins are generally not regarded as durable, while MUF systems with 10-15% of the urea replaced with melamine are considered suitable for moisture resistant grades of both particleboard and MDF. For OSB the resins used are generally PF or PMDI to meet the durability requirement. Most UF bonded products will disintegrate in a boil test, whereas other systems classified as moisture resistant will maintain some integrity. The best performance in the boil test is given by PF systems.

The mode of failure is significant. That adhesive bonded panels swell when exposed to water is clear. That is the expected response from the wood component of the composite. That this swelling is much greater in the direction of the compression applied by the press, in the thickness of the panel, is also clear. However in wood the loss of moisture is accompanied by a return largely to the original dimensions, so that the swelling is regarded as reversible. In the case of adhesive bonded panels the swelling in the thickness direction in particular is not completely reversed when the moisture returns to its original condition. This suggests that some structural change has occurred during the swelling, most likely the breaking of some of the adhesive bonds formed in the press. This is the starting point for the next moisture cycle where the process is repeated, with further bond disruption, ultimately leading to failure.

With the performance of both particleboard and MDF when exposed to cyclic moisture levels problematical, and OSB being less so, it might appear that particle size is significant in the response. Plywood responds to moisture variations in much the same way as does the wood from which it is made. Similarly OSB might be regarded as plywood made using a multitude of very small veneer slices. However the degree of compression necessary to create the level of bonding is higher. The resulting higher density, in terms of the model developed earlier, would be expected to increase the degree of swell. Particleboard and MDF are generally made to higher densities than for OSB, suggesting that the density effect is significant.

The conclusion of this discussion is that there is an inherent inevitability of the wood particles of the panel to expand when moisture content increases, particularly in the panel thickness direction: that is the most significant response of the panel to moisture. The response is influenced to some degree by the process conditions selected, with density, resin type and addition rate being the most significant. The effective limits on these variables for commercial products make the application of particleboard and MDF in exterior exposure situations unlikely to be successful in the increasingly regulated world of building products: but the goal remains. There are major applications for panel products that can be used in exterior applications, but the problem of the disruption and ultimate breakdown of these products in response to moisture changes must be solved in some other way.

There are three approaches that have demonstrated an ability to overcome this limitation. Two are based on modifying the wood, changing the way that it is affected by moisture. The third approach is to encapsulate the wood particles in a matrix that aims to exclude moisture.

Thermal modification is the earliest process. It was developed by Mason (1926) and was used in the first exterior rated product called Masonite. This was a high density fibreboard where the bonding agent was created by heating the wood to a high temperature, above 200°C. This lead to decomposition of the hemicellulosic material, reducing its ability to absorb water. In the Masonite process the high temperature was created in the press; this limited the thickness of the panel that could be made in this way to a maximum of 6 mm. There are other applications of the thermal process; Hsu *et al.* (1989) discussed the improved panel stability achieved by heating the panel after the conventional press in a second pressing stage. Others (Shen, 1988) have described processes for thermal treatment of cellulosic materials before pressing to produce autogenous bonding through thermal modification, while at the same time modifying the swelling response of the material.

The other approach is to modify the cellulose chemically. Rowell *et al.* (1994) discussed the application of both thermal and chemical modification techniques to modify fibre used to manufacture low density fibreboards. The acetylation technique appears to offer significant improvement in the stability of panels, but the technique has not achieved significant commercial success largely because of the cost of the chemicals required.

The third approach is to encapsulate the wood in a material that prevents water reaching the wood. This requires a high level of binder or coating agent. The cement bonded particleboard and fibreboards are examples of this approach, but it also applies to the extruded wood-plastic composite used as an exterior decking in USA. In this product the binder is a thermoplastic comprising 50-60% of the total mass. The wood in this case is also used in a finely divided form, so that the contribution to the structural strength of the product is less than it would be for larger particles.

However particle size is important in the durability performance of these materials. Some wood is always exposed as the material is cut to length or penetrated by fixings. This wood will respond to an increase in moisture by swelling. This is not an issue for small particles isolated within the plastic matrix, but would be more significant with larger particles used in the cement and gypsum matrix products.

9. CONCLUSION

Wood–based panels such as particleboard, MDF and OSB each represent a solution to a particular market demand, often for a product that can be handled like wood but without the limitations of that material. The ability to make panels in a range of sizes and thicknesses, with uniform, predictable properties tailored to particular applications has been a major factor in the significant growth of these materials. Each of these panel types uses a raw material that is either a waste stream from other wood processing operations, or is wood that is unsuitable for other uses. There is thus a ready source of raw material. In addition the processes have demonstrated an ability to use a range of raw materials, both in form, and in different wood species, extending this use to include recycled wood from urban waste streams and to agricultural residues.

This has been made possible by the development of suitable adhesive systems that are able to bond the particles together. The synthetic adhesives offer a consistency of performance that is difficult to achieve with natural products such as tannins, and at a cost that has enabled rapid growth to be achieved. At the same time the adhesive systems have shown a tolerance to a range of wood properties that has enabled most wood residues sources to be used. The environmental effects of formaldehyde emissions from panels made using UF adhesives have been successfully addressed. Initially these required higher resin addition rates to offset a loss in physical property levels, but further development has reclaimed much of this additional cost.

The technology to manufacture these panels has developed rapidly, both incrementally and by breakthroughs. The development of continuous pressing improved costs while offering significant benefits in panel uniformity. This allows large production lines with capacities in excess of 1500 m^3/day for both OSB and MDF, with even higher capacities in particleboard. The plants in Canada, on which the data in Table 12.2 is based, are generally newer plants at the top end of the capacity scale. Unit labour cost at the same wage rates would be higher for smaller plants on a capacity basis, with the benefits of increased automation giving further advantages in newer plants. The result of these developments is a dynamic sector of the wood processing industry, an integral part that contributes to the effective use of the forest resource. It has achieved this by developing new products and the processes that have enabled these to be made efficiently and at reasonable cost.

These processes provide a way of making products over the whole range of densities. While MDF started its life as a panel competing with particleboard as a wood substitute, it has become a new material in its own right. The ability to control density profile so that the surface density of the sanded panel is uniform has provided a consistent response to surface finishing operations. This, together with the thickness tolerance on the panels (typically below ± 0.15 mm) has given a panel that can be processed on automated component manufacturing lines with panels fed to the line, being cut to size, machined, and finished without operator intervention.

The MDF process is now used to produce panels well outside the density range envisaged as medium density. Low density panels in the 500-600 kg/m^3 range are now common in the market, while higher density products are produced, specifically as a substrate for prefinished flooring systems. The MDF process is also used to produce the fibre for moulded doorskin products. Thin MDF (<6 mm) became economic with continuous presses, firstly with the Mende design and subsequently in the straight through presses. Thin MDF grew rapidly as an alternative to hardwood plywood at a time when the availability of this material was declining. It is perhaps appropriate to redefine MDF as process rather than a product, one that can produce an adhesive-fibre mix that can be used to produce a whole range of fibre bonded products.

Particleboard has also responded to the MDF challenge to its market area by improving performance in a number of ways. The density profile is now monitored and controlled and the importance of particle geometry in determining panel characteristics is recognised. The fixing characteristics of particleboard are not as good as for MDF but new designs of fittings have overcome much of this limitation. In another direction the possibility of using larger particles to provide a structural particleboard that reaches into the OSB market is being actively considered.

OSB was developed with a single focus, as an alternative to softwood plywood for sheathing houses, but is being used for a wide range of tasks. Many of these currently use plywood, but the cost benefit of OSB and its demonstrated performance make it an alternative that must be considered in a competitive market.

The overall picture is of the wood–based panels sector continuing to grow as products are developed that meet new customer requirements. The processes have demonstrated an adaptability to different fibre sources, both in form and in species that has allowed plants to be established in most parts of the world, primarily to meet local demand, but also with a significant export component to those areas where fibre supplies are fully committed.